



Technical Occupational Welding

Learner's Guide

Year 2

TERM 2

Re-Cap Safety and apply work site practices from term 1 work.

Equipment, Hand tools & Power Hand Tools

Hand tools examples:

Welding clamps

Welding Earth Clamp- Croc-Type 	Welding earth clamp G-type 600 Amp 
welding electrode holder 	welding cable kit 
 VICEGRIP WELDING CLAMP 230MM Locking pliers are equipped with a hex adjusting screw on the adjustment bolt Adds an additional controlled release of the tool and allows fine adjustment. This provides an ideal solution for the accurate alignment of both round and square shapes. The U shaped jaws allow for the visibility of the weld area during welding.	

BOLT CUTTERS

What is the use of bolt cutter?

A bolt cutter, sometimes called bolt cropper, is a tool used for cutting bolts, chains, padlocks and wire mesh. It typically has long handles and short blades, with compound hinges to maximize leverage and cutting force.



Spanners & Sockets

What are Spanner Sets?

[Spanner sets](#) consist of a selection of different spanner sizes, bought together as a single kit for tackling nuts and bolts of any size.

What are Spanner Sets used for?

Spanners are used for loosening or tightening nuts and bolts.

Different types of Spanner Sets

Nearly all ratchet-type spanner sets will allow this feature to be deployed in both directions, either by flipping the spanner over or by adjusting a small lever on the handle or head of the unit.



8 Pcs 72 Teeth Ratcheting Spanner Set

Usually sold in open-ended, flare or ring head configurations.



20 PCS Ratchet Wrench Box

Ring Spanner Sets The potential downside of ring spanner sets is that the tools must be lifted and replaced on the fastener once the full turning arc has been exhausted. 	 Box Spanner Sets The elongated shape of box spanners makes them very handy for accessing nuts that are deeply recessed or otherwise in difficult-to-reach places.
	Socket and spanner sets very often feature a ratchet system, meaning they don't have to be lifted off the nut between turns

Imperial Spanner Sets are defined by imperial measurements (inches and fractions of inches).

Metric Spanner Sets are designed to tackle nuts and fasteners defined by millimetre sizing (as opposed to imperial inch-based kits for older fittings).



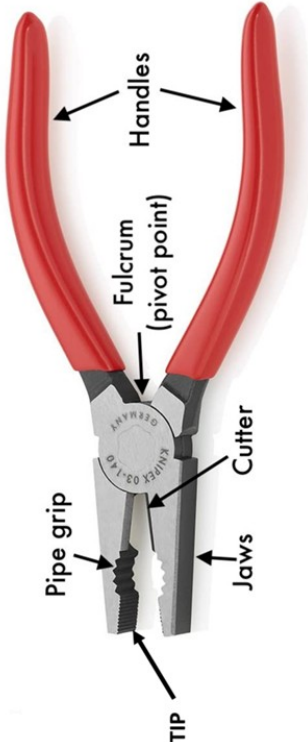
Adjustable Spanner Sets

Adjustable spanners is a standard layout open-ended head design, but with

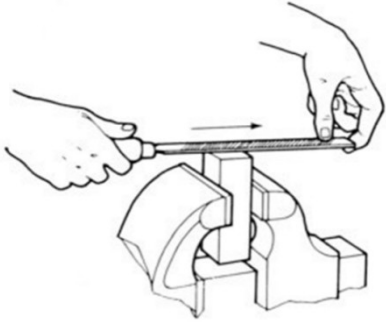
An additional function that allows for manual opening or closing of the

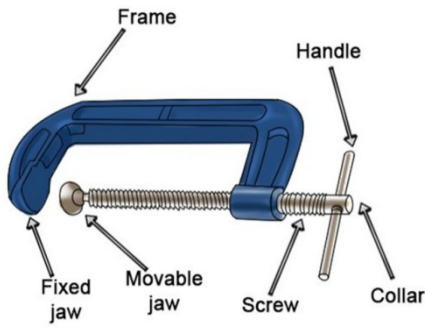
Spanner's jaw via a rotating cylinder driven by a screw. The mechanism

enables a single spanner to tackle numerous gauges of nuts and fasteners.

PLIERS		
	Handles	These are used to hold onto the pliers Some pliers have insulated handles so the pliers can be used on electrical wiring.
	Fulcrum:	This is the pivot point around which the parts of the pliers rotate.
	Jaws:	The jaws has a specific size and determines the size of the pliers.
	Pipe Grip	The rounded shape of the pipe grip allows you to hold on tight to rounded objects, such as pipes.
	Tip	The tip is used to pinch or hold an object with precision

VICE GRIP		
	Screw:	The screw is used to adjust the jaw opening
	Handle/s:	The handles are used to close and lock the vise-grip into position.
	Spring:	This helps the jaws open when the release lever is activated
	Link:	The link is the locking mechanism of the vise-grip.
	Grip:	The grip is manoeuvred with the opposing fingers of the hand to press

FILES	
	
Clamp the work piece securely in a bench vice.	
Ensure that you have scribed the target line clearly onto the work surface on the operator side.	
Stand squarely in front of the workpiece with your feet slightly apart, the dominant foot slightly in front of the other.	
Using your dominant hand on the handle and the other hand on the tip to steer the file, press down slightly and file away from your body.	
Slightly lift the file, retract and repeat. Make sure you inspect the cutting edge of the file and the target line regularly and adapt your file and body position to correct the cut.	

G-CLAMP	
	
Frame:	The frame is made of steel and is the backbone of the g-clamp.
Fixed Jaw:	This is the anvil against which the workpiece is placed.
Screw:	The working piece of the G-Clamp, this will close or open the gap when adjusted.
Movable Jaw	This anvil is attached to the screw and clamps the workpiece when the

	screw is tightened.
Handle:	A T-handle helps tighten the screw.
Collar:	The end of the screw that houses the T-handle.

POWER TOOLS

How to replace a switch and carbon brushes

DIE GRINDER

A die grinder is a powerful handheld tool used to **grind, sand, hone, or polish materials**. You can use it for various tasks, including metalworking, woodworking, plastics, and auto body care.



What is SDS rotary drill?

SDS is short for Slotted Drive Shaft or Slotted Drive System.

These drills are typically used for heavy-duty drilling into materials such as brick, stone, concrete and other types of masonry. Unlike regular drills, they have a unique design that provides a high level of power during drilling.



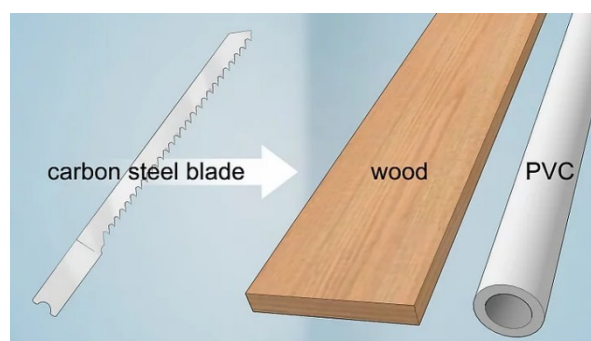
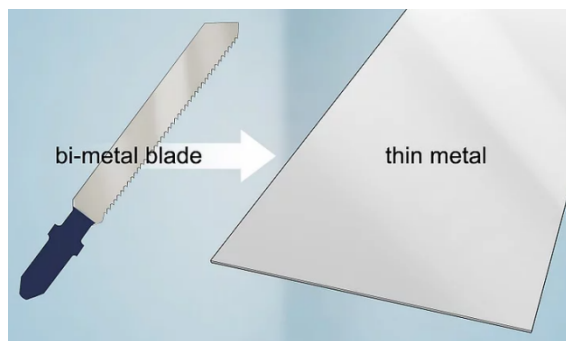
What's the difference between an SDS and rotary drill?

SDS drills combine the standard rotary action and hammer action of a standard drill but use different drills and typically have more power. They have a chuck that is specially designed to fit SDS drill bits.

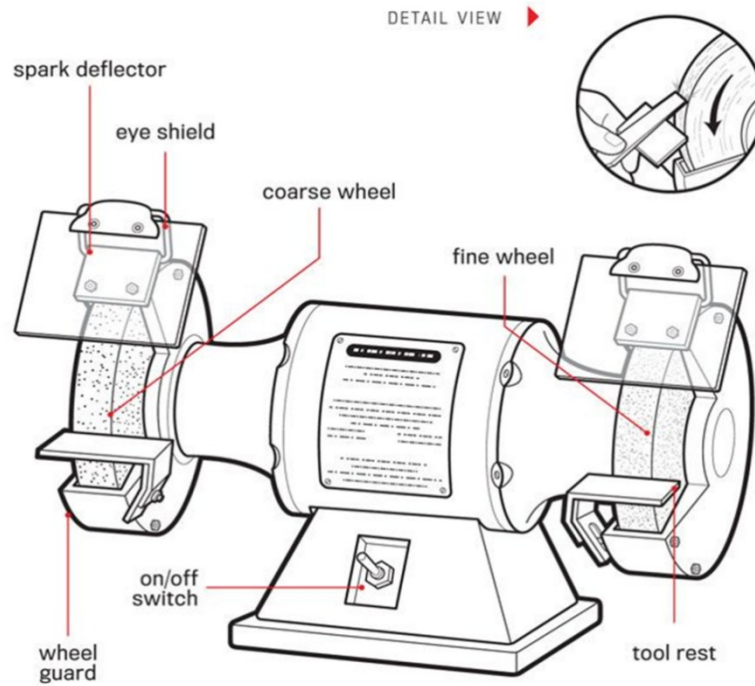
JIG SAW



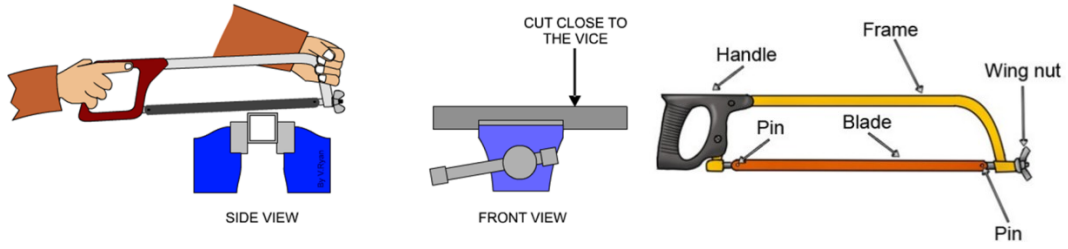
A jigsaw is one of the most versatile power tools you can have in your workshop. It can cut through materials such as wood, metal, laminate, and PVC, as well as make straight and curved cuts easily. When working with a jigsaw, make sure to use the correct blade for the material you're cutting and wear safety gear.



BENCH GRINDER



On/Off Switch:	Activates the Bench Grinder
Wheel Guard:	Protects the grinder wheel against objects bumping into the grinder
Course Wheel:	Makes rough grinds and removes large portions of material but gives a rough finish
Spark Deflector and eye shield:	Protects the user from flying sparks that emit from the wheel during use.
Fine wheel:	Removes less material at a time, but leaves a finer more accurate cut.
Tool Rest:	For steadying the work piece.

HACKSAW	
 The diagrams illustrate the use and components of a hacksaw. The first diagram shows a side view of a person using a hacksaw to cut a piece of material held in a blue vise. The second diagram shows a front view of the same setup, with an arrow pointing to the cut line labeled 'CUT CLOSE TO THE VICE'. The third diagram is a detailed view of the hacksaw's components, labeled: Handle, Frame, Blade, Wing nut, and Pin.	
Pins	The pins hold the hack saw blade in place.
Blade:	Made from strong steel blades come in different sizes or teeth per inch.
Handle:	This is shaped to fit your hand and you will take this part in your dominant hand.
Frame:	The extension of the handle to help the blade's full length fit to the saw.
Wing Nut:	The wing nut tightens the blade and should be relaxed between uses to the blade does not stretch out or the frame becomes buckled. The wing nut is important because the blade needs to be taut.

Locating and Removing the Old Brushes

The brushes are small, carbon blocks that help conduct electricity from the power source to the rotating parts of the motor.

Over time, these brushes wear down and become less efficient, causing the motor to run poorly or not at all.

To remove the old brushes.

First disconnect the motor from its power source and remove any protective covers.

Locate the position of the brushes inside the motor and gently ease them out with a pair of pliers, taking care not to damage the armature or commutator.

It's important to make note of the direction the brushes are facing before removing them to ensure proper replacement.

Once the old brushes are out, clean the brush holders and commutator to prepare for the installation of new brushes.

Installing New Brushes

First, make sure you have the correct replacement brushes for your specific drill model.

Insert the new brushes and push them in until they are secure.

Finally, replace the brush caps and test out your newly replaced brushes.

Keep in mind that brushes wear out over time and will need to be replaced periodically to keep your cordless drill running smoothly.

Test and Replace the Motor Cover

Before you even start replacing the brushes, it's important to test and replace the motor cover if required. A drill motor cover is designed to protect the armature and other essential parts of the motor.

<https://www.youtube.com/watch?v=oEMJfhLf9f0>

<https://www.youtube.com/watch?v=5qlwBrUhpH8>

<https://www.youtube.com/watch?v=qCyTDAsPCZQ>

<https://www.youtube.com/watch?v=4WE4v5dlswQ>

Replace the Switch on a Cordless Drill



Dead power drill?

Replace the tool's power switch.

One of the most common power tool repairs is replacing its switch. The switch is part of the tool's electrical system and may start to wear out after a few years of use.

A broken switch can also cause the drill to lose variable speeds or fail to switch between forward and reverse mode.

To diagnose and repair a drill switch, you'll need a screwdriver, needle-nose pliers and a multimeter.

This is how you start

OPEN THE DRILL CASING



Remove the transmission screws.

Unscrew the two screws holding the transmission to the case, then a series of screws holding the two case halves together.

Separate handle assembly.

Once that's done, separate the two halves of the drill. If there's a sticker on the base holding the two halves together, you'll need to cut through it with a box cutter.



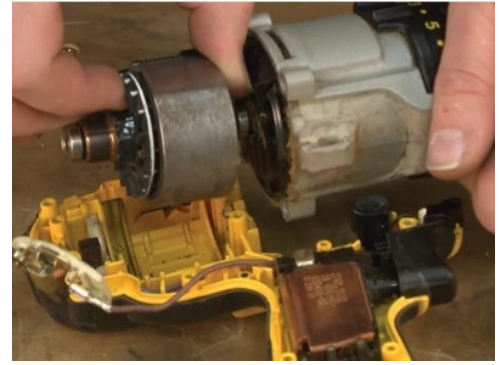
Remove bearing caps.

Next, in order to remove the switch, you'll need to remove the wires connecting the switch to the brush holder.



Remove brushes.

Now you're able to separate the brush holders from the wires going to the switch. Be sure to make note of which wire goes to which brush holder, as you'll need to reassemble it the same way.



Pull motor and transmission assembly from drill.

Remove the drill's motor assembly.

Remove the motor by removing the two screws on the outside the drill case that hold the transmission in place.

Remove the two blocks holding the motor assembly in place.

These blocks are attached to the case by four screws.

Once those are removed, you're free to pull the motor and transmission assembly away from the drill



Remove switch.



Remove the switch from the drill.

The wires of the switch can be tested with a multimeter.

Test the current on both switches.

Test its electrical continuity with a multimeter

Set your multimeter to the continuity setting.

Checking for continuity is done in the ohm's settings.

An ohm is the unit of measurement of resistance in a circuit.

Many multimeters have an audible tone setting which can be used when checking for continuity.

Touch one of the test probes on the multimeter to one of the battery connections on the switch.

Then touch the other test probe to the opposite connection and pull the switch trigger.

Repeat this process on the other side.

If your multimeter does not beep or otherwise indicate resistance on one or both sides, you'll need to replace the drill's switch.



Set the new switch and Test the new switch with the multimeter before installing.

Route wires into housing.

Place the new switch into the drill body where the old one was. Route the switch's wires back into the drill's housing.



Replace the wire to the lower brush assembly and replace the assembly back into the drill.

Replace brush and secure wiring

Replace motor and transmission assembly from drill.

Reinstall the motor and transmission assembly back into the drill, and replace the blocks around the armature bearings to secure it to the housing.



Reattach and replace top brush.

Replace the two screws on the outside of the case that hold the transmission in place.

Reconnect the wire to the second brush holder and reinstall it into the drill. Tuck any wires that come out of the casing as necessary.

Replace remaining transmission screws.

Finally, reinstall the other half of the drill casing, screw it together and install the last two transmission screws.

Your drill should now be fully operable,

Storage and Cleaning Hand Power Tools

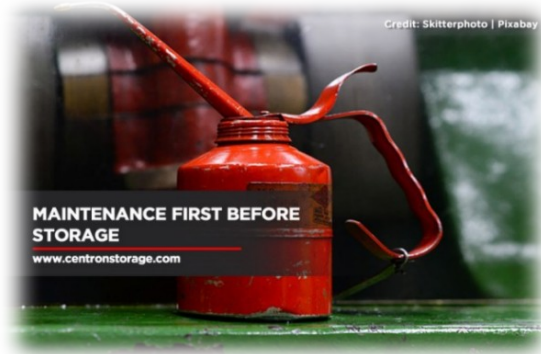
WAYS TO SAFELY STORE YOUR POWER TOOLS

Power tools are made up of **complex mechanical and electrical parts**, so it's essential that they **require maintenance before you place them in storage**.

This ensures that your power tools will work perfectly the next time you use them.

THE BEST PRACTICES FOR PRE-STORAGE POWER TOOL MAINTENANCE:

- Clean Your Power Tools After Every Job



Dust, debris, and moisture build-up can cause mechanical and electrical failure. So before you store your power tools, make sure you clean them with a rag or old towel after every use.

- Check for Damage and Dispose of Defective Tools

Power tools are technically electrical devices so they should be checked for any damage before storage.

- Make Sure Batteries Are in Good Condition



Cordless power tools rely on their batteries to function.

For lithium-ion batteries (Li-Ion), just make sure to store them with at least a 50% battery charge.

Lubricate Your Power Tools

Lubricating your power & hand tools helps **prevent rust** and ensures that moving parts don't get stuck and function as intended. Check the manual for each power tool **for the areas to be lubricated** as well as the type of lubricant to be used.

- Remove Rust From Your Tools

If your power tools such as drills show signs of rust, it is highly recommended that **you must remove them right away**. **Rust tends to spread all over the tool with time** and **the corrosion and abrasion damage can cause parts to fail, freeze up, or cause structural damage**.

STORAGE SOLUTIONS FOR TOOLS

CHOOSE THE RIGHT AREA FOR STORING YOUR TOOLS

When choosing a dedicated area to store your tools, one of your primary goals to reduce any negative environmental effects.

Here are the things to consider when choosing a storage area for your tools:

- a) An area with less exposure to sunlight and heat.
- b) Avoid areas with **high relative humidity**.
- c) Choose an area with low foot traffic to avoid people accidentally knocking over your power tools and cause injuries.
- d) A climate-controlled room is recommended.

RECOMMENDED AREAS TO STORE YOUR POWER & HAND TOOLS:-

- a. The garage can be used as a storage place.
- b. Have a dedicated tool shed built if you have enough space outside your house.
- c. For schools have a storeroom.

1. KEEP THEM IN TOOL-STORAGE CABINET

Benefits to keeping your tools in their own cabinet:

- It helps keep your tools organized and secure.
- The cabinet doors help protect your tools from the elements as well as physical damage.
- Some power tool storage cabinets come with charging stations for your cordless equipment.

2. PLACE YOUR EQUIPMENT ON POWER TOOL STORAGE RACKS

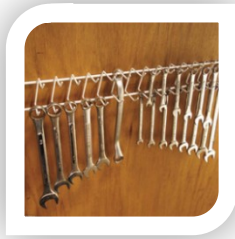


3. USE THE ORIGINAL PACKAGING.

The original packaging has built-in internal supports that were designed to secure and protect an item from impact damage.

4. HANG YOUR POWER & hand TOOLS

A cost-effective storage solution for your tools is to hang them on pegs on walls or shelves.



5. STORE POWER TOOLS INSIDE CLEAR PLASTIC CONTAINERS

These boxes have a wide range of benefits, such as:

- Plastic containers are more durable than standard cardboard boxes.
- Most clear plastic boxes come with lids, so they keep dust and debris away from your equipment.
- Plastic containers are resistant to most liquids, so your power tools are safe from accidental spills.
- Clear Plastic boxes make it easier for you to find your power tools



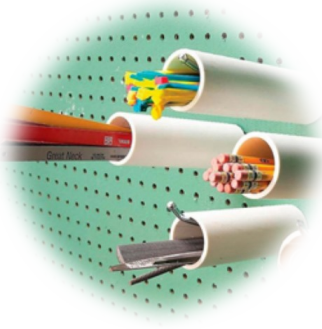
Toolbox Liner

Ruler Tidier Tool Trays look great in toolboxes, but there's a cheaper alternative. Cut a nonslip rug mat to fit any size drawer and keep tools from sliding around



Corner-on Pegboard Hooks

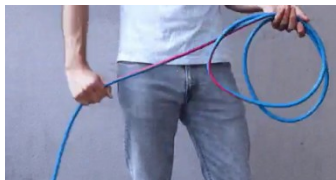
Pegboard Cubbyholes



PROPER CARE AND STORAGE OF EXTENSION LEADS

If you take good care of your extension cords, they should last for many years.

The over-under coil



- <https://cdn.thewirecutter.com/wp-content/uploads/2018/08/over-under-570.gif?format=mp4>

❖ Video: Kyle Fitzgerald

- **Pros:** Won't damage the cord, leaves it ready to use next time

Cons: The method doesn't come naturally, so you'll need to get used to it

- A regular coil, making loop after loop on top of one another, actually bends an extension cord out of its natural curve, making it more cumbersome to handle over time and stressing the wires inside.
- Velcro cord wraps



<https://cdn.thewirecutter.com/wp-content/uploads/2018/08/velcro-570.gif?format=mp4>

<https://youtu.be/tcOhVeWiVY4>

Video: Kyle Fitzgerald

The chain-link

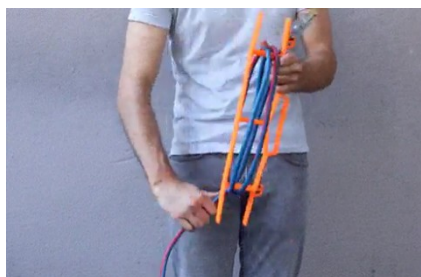
<https://cdn.thewirecutter.com/wp-content/media/2021/01/daisy-chain-sequence-570.gif?format=mp4>



Fold the cord in half, and loop the half end through like a loose knot. Video: Kyle Fitzgerald

Leaves cord ready to use, nothing to buy and the chain-link method takes a little practice, does a nice job of keeping your cord organized and easy to transport for next time.

Spools

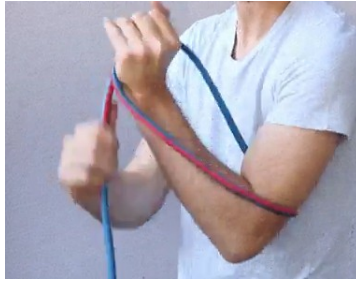


<https://cdn.thewirecutter.com/wp-content/uploads/2018/08/spool-570.gif?format=mp4>

Video: Kyle Fitzgerald

Requires unspooling cord before use, bad for the cord, and a waste of a few Rands that could be better spent on anything else

The around-the-arm coil



<https://cdn.thewirecutter.com/wp-content/uploads/2018/08/arm-wrap-570.gif?format=mp4>

Video: Kyle Fitzgerald

Pros: Quick and easy, nothing to buy

Cons: May damage the cord over time, often results in tangles, and cord is not easy to use the next time



Material Specifications means the description of the Material, including requirements, tolerances, shelf life, specifications, suppliers and safety data and testing instructions.

Identification process to identify materials to specification on:-

Ferrous metals include:

- Mild steel
- Carbon steel
- Stainless steel
- Cast iron
- Wrought iron

Metal Identification Tests Guide

Common Metal Testing Methods

1. Magnetic Tests

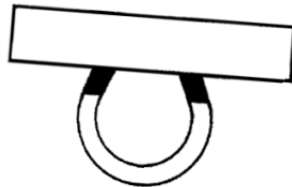
Magnets are frequently used for metal identification. Ferrous iron-based alloys are magnetic, while nonferrous metal is non-magnetic.

Magnetic Metals

If a metal clings to a magnet, it means that it is ferritic.

Strongly magnetic materials include:

- Types of Steel
 - Carbon steel
 - Low-alloy steel
 - stainless steels

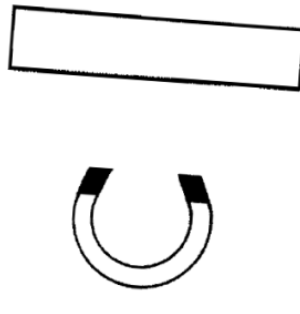


A Magnet Clinging to Metal Indicates a Ferritic Metal

NON-MAGNETIC METALS

Nonmagnetic materials include:

- Copper-base alloys
- Aluminium-based alloys



Non-magnetic Steel is austenitic

2. Metal Spark Test

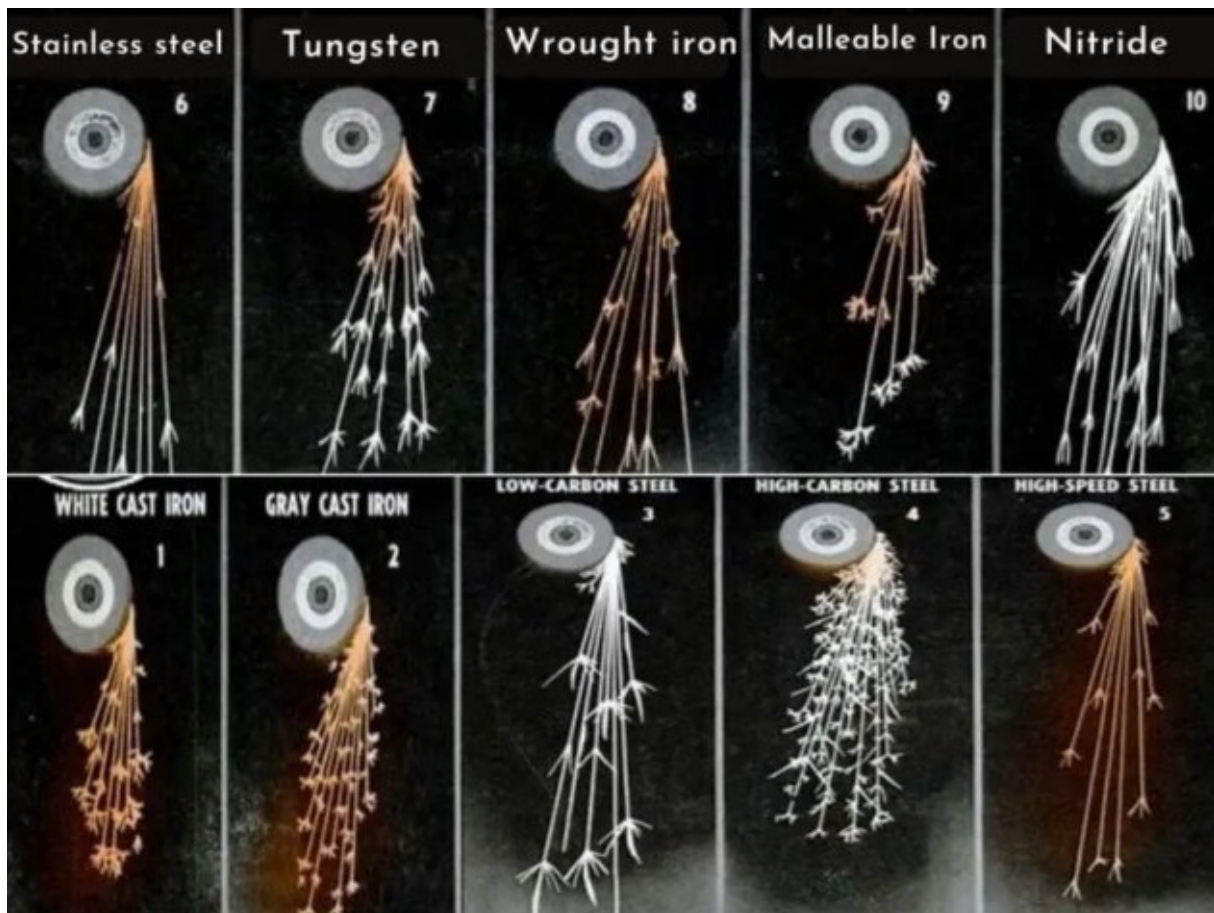
What is a spark test?

Spark testing is a method of determining the general classification of ferrous materials.

It normally entails taking a piece of metal, usually scrap, and applying it to a grinding wheel in order to observe the sparks emitted

Take note and visually inspecting the spark colour, shape and length.

When examining a “carrier line” look at the spark length, stream, and colour.



Spark Characteristics

- **Wrought iron:** Wrought iron sparks flow out in straight lines. The tails of the sparks widen out near the end, similar to a leaf.
- **Mild steel:** Mild steel sparks are similar to wrought irons; except they will have tiny forks and their lengths will vary more. The sparks will be white in colour
- **Medium-carbon steel:** This steel has more forking than mild steel and a wide variety of spark lengths, with more near the grinding wheel.
- **High-carbon steel:** High-carbon steel has a bushy spark pattern (much forking) that starts at the grinding wheel. The sparks are not as bright as the medium-carbon steel ones.
- **High-speed steel:** High-speed steel has a faint red spark that sparks at the tip.
- **Stainless steel:** These sparks are not so dense as the carbon steel sparks, do not fork, and are orange to straw in colour.
- **Cast iron:** Cast iron has very short sparks that begin at the grinding wheel.

Steps in the identification of materials to specification to be in logical sequence and to job requirements.

The correct application for the following materials:

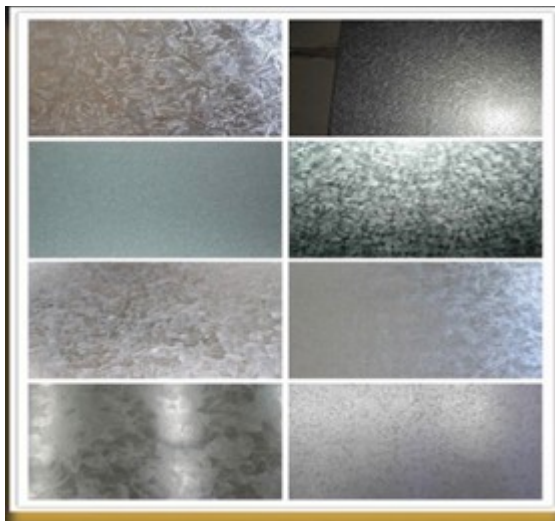
Galvanised materials



Low initial cost compared to most treated steels. Galvanized steel is immediately ready to use when delivered.

Longer life. With galvanization, a piece of industrial steel is expected to last more than 50 years in average environments and can last over 20 years with severe water exposure.

Rust resistance from the zinc coating. Galvanized steel is very protective, including sharp corners and recesses that couldn't be protected with other coatings, making it resistant to damage.



What industries utilize galvanized steel?

Galvanized metals are used everywhere!

The bodies of cars and many bicycles are made from galvanized metals.

Some drinking water pipes are still made from galvanized steel.

Cool rolled sheet metal is also frequently galvanized.

Nuts, bolts, tools, and wires of all kinds are now galvanized because it is a cheap process, and helps boost the metal's lifespan!

How to store mild steel and galvanized materials

Mild steel rusts so when two (or more) dissimilar metals are brought into electrical contact under water and a galvanic couple forms then one of the metals in the couple becomes the anode and corrodes faster than it would all by itself, while the other becomes the cathode and corrodes slower than it would alone.



Types of Mild Steel

Mild steel comes in a variety of types to suit different applications. The most common are:

Low-carbon steel

Contains less than 0.3% carbon. It's cheap, ductile and workable, but not very hard, making it useful for manufacturing bolts, nuts, and structural shapes.

Medium-carbon steel

Has 0.3-0.6% carbon. Stronger than low-carbon steel and suitable for shafts, gears, and axles.

High-carbon steel

Contains 0.6-1.0% carbon. Very hard, but brittle, used for cutting tools, springs, and knives.

Grades of Mild Steel

Mild steel also comes in various grades, categorised on its chemical composition. This determines its strength, durability, and malleability. Some of the most common types on the market are:

A36

A36 is a low-carbon steel with good strength and formability. It has a carbon content of 0.25% and a tensile strength of 400-550 MPa. Its structural steel is used for beams, plates and sheets in the construction industry. This versatile mild steel is suitable for welding and can withstand high heat.

S275JR

S275JR has a carbon content of 0.2% and a tensile strength of 410 MPa. It is a non-alloy structural steel used in a variety of welded, bolted and riveted constructions. S275JR is mainly used for bridges, buildings and general engineering structural applications. This mild steel grade is easily weldable and machinable.

S355JR

S355JR has a carbon content of 0.24% and a tensile strength of 470-630 MPa. It is a high-strength, low-alloy structural steel used in a variety of welded, bolted and riveted components, such as bridges, overhead cranes and buildings. S355JR provides high yield and tensile strength with good formability and weldability.

Other common mild steel grades are Q195, Q215, Q235B and Q345B. The Q refers to the yield strength, 195 means the yield strength is 195 MPa. These Chinese steel grades have similar properties and applications as the EN standards like S275JR and S355JR.

Uses and Applications of Mild Steel

Construction

Mild steel provides structural support for large buildings through steel beams and reinforcement for concrete through steel rods; also corrugated sheet metal made of mild steel serves as roofing.

Transportation

Serving as an indispensable material in the transportation industry, mild steel constitutes the ships, trains, cars, and aircraft that move people and goods around the world; specifically, the valued strength-to-weight ratio of mild steel alloys allows vehicles to achieve an optimal balance of lightness and sturdiness through the use of steel plates, beams, chassis, and frames.

Tools & Equipment

The versatility and affordability of mild steel make it a vital component in many common tools, appliances, and pieces of equipment, ranging from wrenches and pliers to ovens and refrigerators to diverse industrial machinery; its ability to withstand high heat and physical stresses while being readily shaped into specific forms lends it well to these functions.

Containers

Capitalizing on mild steel's capacity to maintain its form, containers of all varieties including drums, tanks, bins, and cans rely on this material to store and transport liquids, powders, granules, and other contents securely in factories, kitchens, and everywhere in between.

The Most Common Uses of Stainless Steel include:

- Architecture and Construction
- Automotive and Transportation
- Medical

- Energy and Heavy Industries
- Food and Catering

Aluminium

Uses of Aluminium

The typical uses of aluminum are listed below with explanations.

Foil



Food wrapping is one of the most common uses of aluminium foil. It is aluminium in the form of thin metal leaves.

Cans



Aluminium cans store various food and beverage products, such as soda, beer, and meals. Cans made of aluminium are also recyclable, making them a sustainable choice.

Foil Containers



Gutters

Windows



Aluminium windows are a durable, adaptable, and lightweight option for your home's windows. Along with their durability and low maintenance needs, aluminium windows are an effective option for new construction and remodelling projects.

BRASS

Brass's resistance to corrosion and bacteria makes it an ideal material for plumbing applications. From faucets to valves, brass ensures clean water flow and durability, making it a staple in plumbing systems worldwide.

Brass's high strength, corrosion resistance, and low friction properties make it a preferred choice for various machinery and equipment. It is commonly used in bearings, gears, valves, and other components that require durability, precision,

Uses for Brass:

Door handles and knobs:

Musical instruments:

Jewelry and accessories:

Decorative items and artwork:

Coins and currency:

Zippers and fasteners:

Keychains and key rings:

Lamps and lighting fixtures:

Uses for Brass in Industry:

Construction and architecture

Automotive industry

Electrical and electronics industry:

Machinery and equipment:

Answering Common Questions:

1. What are the common uses of brass?

Brass finds its way into a multitude of everyday items, including door handles, faucets, musical instruments, jewellery, decorative items, coins, zippers, keychains, lamps, and hinges, just to name a few. Its versatility, durability, and aesthetic appeal make it an integral part of our lives.

2. How is brass used in plumbing and electrical work?

Brass's corrosion resistance, durability, and excellent conductivity make it an ideal material for plumbing fittings, valves, and connectors. In electrical work, brass is used for connectors, terminals, circuit boards, socket pins, and shielding against electromagnetic interference (EMI).

3. Can brass be used for decorative purposes?

Absolutely! Brass's warm golden hue, malleability, and timeless elegance make it a popular choice for decorative items, artwork, and jewellery. Its ability to be shaped into intricate designs adds a touch of sophistication and charm to any space.

4. What musical instruments are made from brass?

Brass is a key player in the world of music. Instruments like trumpets, trombones, saxophones, tubas, and French horns are typically [made from brass](#) alloys. The unique acoustic properties of brass give these instruments their rich, vibrant sound.

5. Are there any unique or unexpected uses of brass?

Indeed! Brass's versatility has led to some unexpected applications. It is used in fire sprinkler systems for its corrosion resistance, in coinage for its distinctive appearance, and in heat sinks to dissipate thermal energy. Brass's combination of properties opens up possibilities across various industries.

WELDING

Welding symbols and joints

- Sketches of weld symbols can be done free hand or formal drawing format.

Joining Metals

Welding is a high heat process which melts the base material, with the addition of a filler material.

Heat at a high temperature causes a weld pool of molten material which cools to form the join, which can be stronger than the parent metal.

Welding symbols used in the industry

Weld symbols are a very useful way of communicating welding requirements from the design office to the shop floor.

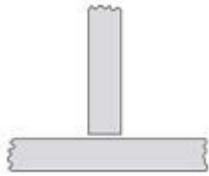
Common Joint Configurations

Butt Joint



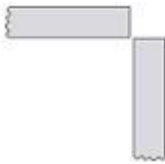
A connection between the ends or edges of two parts making an angle to one another of 135-180° inclusive in the region of the joint.

T Joint



A connection between the end or edge of one part and the face of the other part, the parts making an angle to one another of more than 5° up to and including 90° in the region of the joint.

Corner Joint



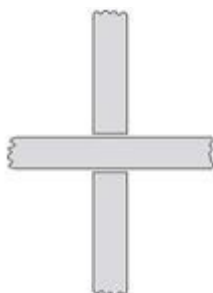
A connection between the ends or edges of two parts making an angle to one another of more than 30° but less than 135° in the region of the joint.

Edge Joint



A connection between the edges of two parts making an angle to one another of 0 to 30° inclusive in the region of the joint.

Cruciform Joint



A connection in which two flat plates or two bars are welded to another flat plate at right angles and on the same axis.

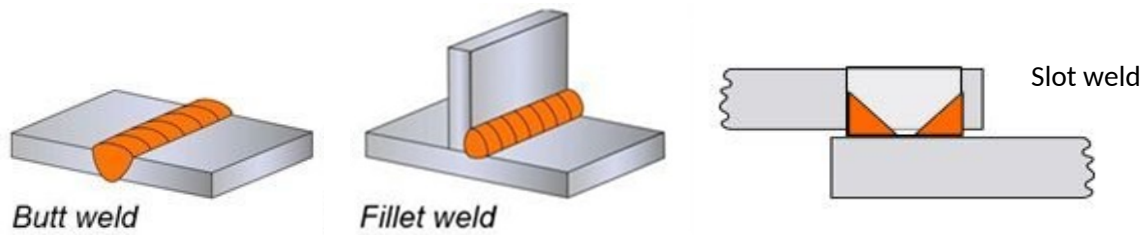
Lap Joint



A connection between two overlapping parts making an angle to one another of 0 - 5° inclusive in the region of the weld or welds.

Types of Welding Joints

Welds Based on Configuration



Joint between two overlapping components made by depositing a fillet weld around the periphery of a hole in one component so as to join it to the surface of the other component exposed through the hole.

Based on Penetration

Full penetration weld



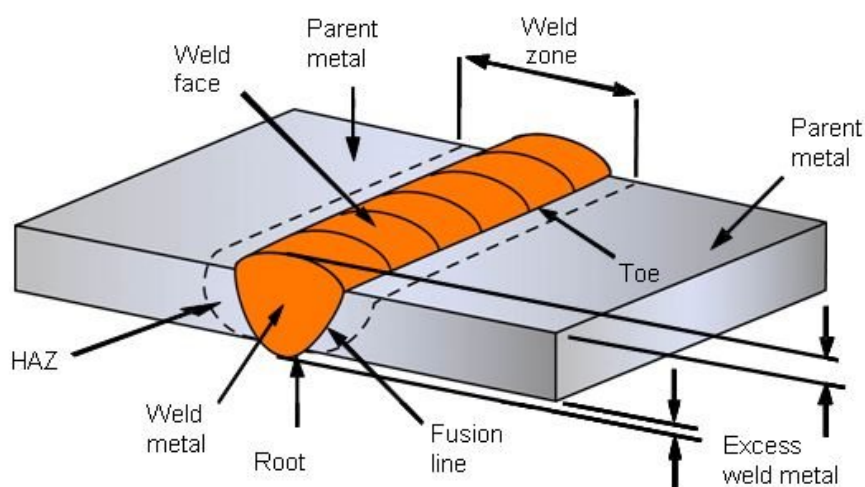
Welded joint where the weld metal fully penetrates the joint with complete root fusion. See AWS D1.1).

Partial penetration weld

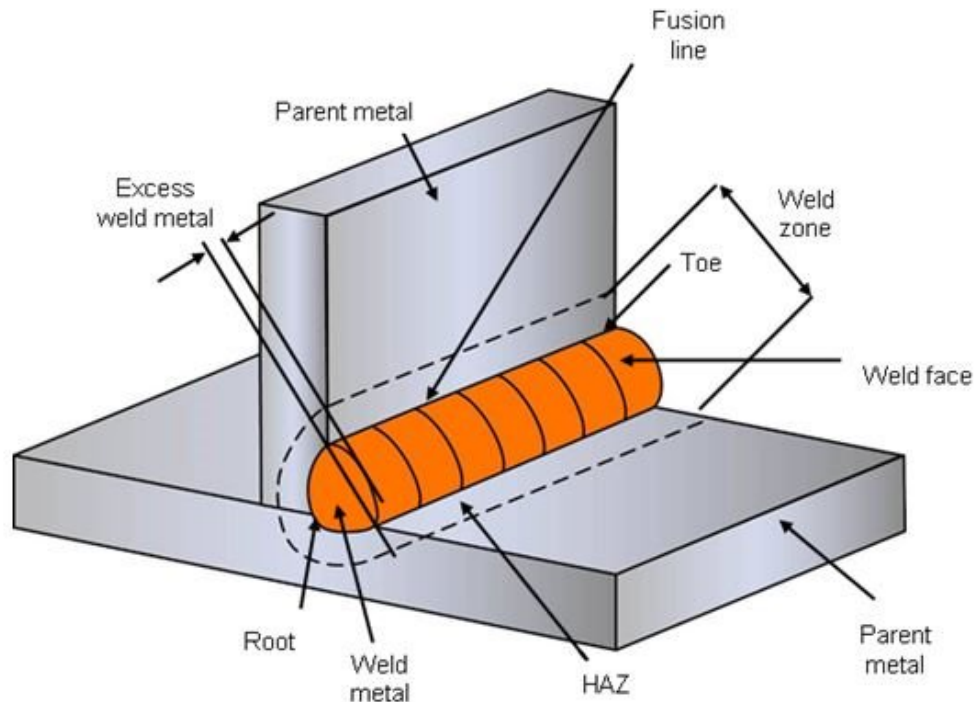


Weld in which the fusion penetration is intentionally less than full penetration. This is referred as preferred term is partial joint penetration weld (PJP).

Features of Completed Welds



Butt weld



Fillet weld

Welding joints and their terminology

Parent / Base Metal

Metal to be joined or surfaced by welding, braze welding or brazing.

Filler Metal

Metal added during welding, braze welding, brazing or surfacing.

Weld Metal

All metal melted during the making of a weld and retained in the weld.

Heat Affected Zone (HAZ)

The part of the parent metal metallurgically affected by the weld or thermal cutting heat, but not melted.

Fusion Line

Boundary between the weld metal and the HAZ in a fusion weld. This is a non-standard term for weld junction.

Weld Zone

A zone containing the weld metal and the HAZ.

Weld Face

The surface of a fusion weld is exposed on the side from which the weld has been made.

Weld Root

Zone on the side of the first run furthest from the welder.

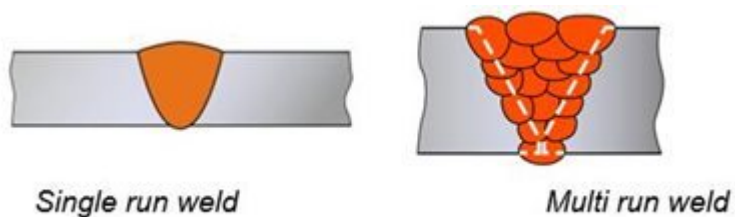
Weld Toe

Boundary between a weld face and the parent metal or between runs. This is a very important feature of a weld since toes are points of high-stress concentration and often they are initiation points for different types of cracks (e.g. fatigue cracks, cold cracks).

Excess Weld Metal

Weld metal lying outside the plane joining the toes. Other non-standard terms for this feature: are reinforcement, and overfill.

Run (pass)



The metal melted or deposited during one passage of an electrode, torch or blowpipe.

Layer

Stratum of weld metal consisting of one or more runs.

Welding symbols in accordance with AWS and work instruction sheet.

Basic Requirements

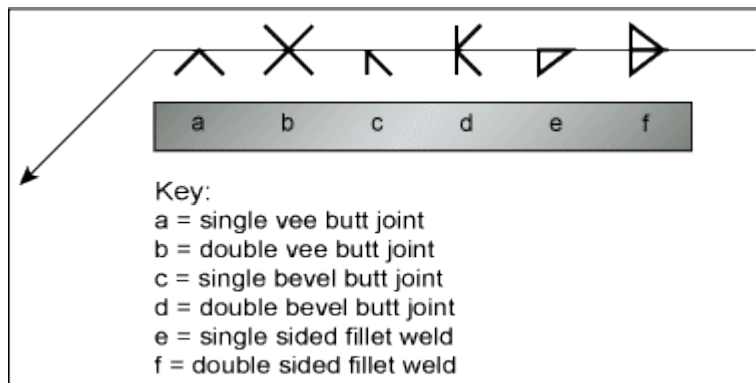
All the standards have the same requirements in relation to the following items:

- Arrow line and arrowhead
- Reference line

The arrow line can be at any angle (except 180 degrees) and can point up or down. The arrowhead must touch the surfaces of the components to be joined and the location of the weld.

The symbols, in particular for arc and gas welding, are often shown as cross-

sectional representations of either a joint design or a completed weld. Simple, single edge preparations are shown below.



Butt welded configurations would normally be shown as a convex profile (Fig.7 'a', 'd' and 'f') or as a dressed-off weld as shown in 'b' and 'c'. Fillet weld symbols are always shown as a 'mitre' fillet weld (a right-angled triangle) and a convex or concave profile can be superimposed over the original symbol's mitre shape. See Fig. 7.

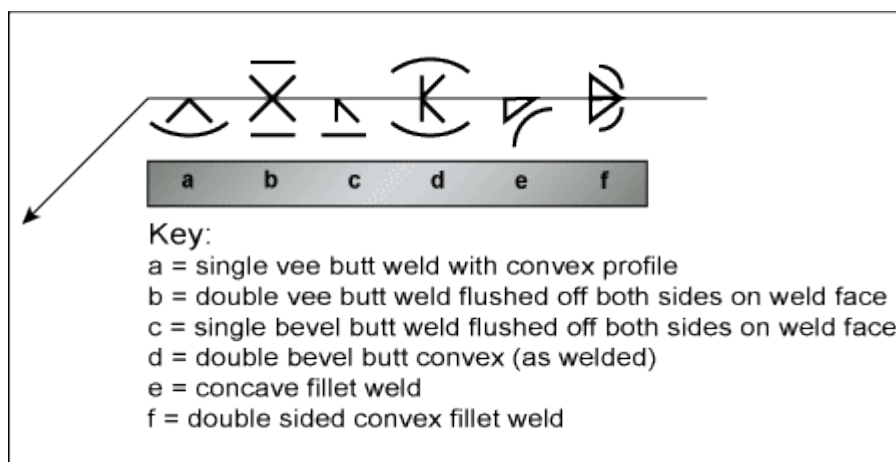


Fig. 7.

For butt joints and welds, an S with a number to the left of a symbol refers to the depth of penetration as shown in Fig.9.

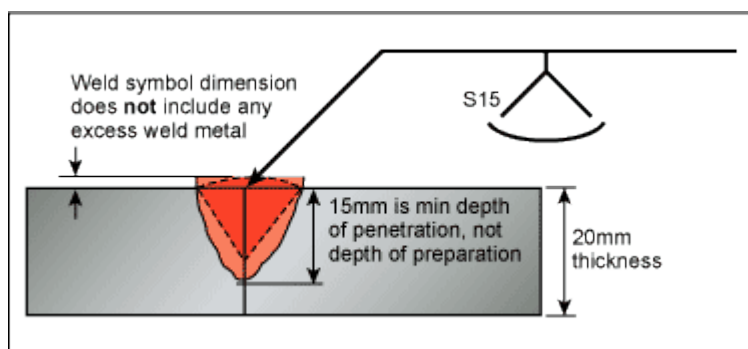


Fig.9

When there are no specific dimensional requirements specified for butt welds on a drawing using weld symbols, it would normally be assumed that the requirement is for a full penetration butt weld (Fig.10).

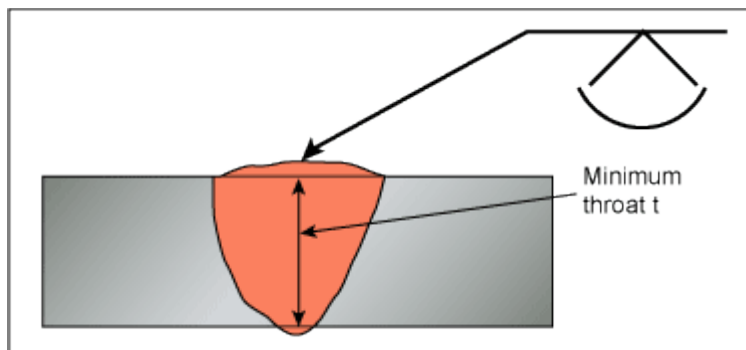


Fig.10

Welding symbols provide a system for placing welding information on drawings and work sites for the purpose of relaying information to fitters, welders, fabricators, inspectors, etc.

These symbols quickly indicate the type of weld joint needed to satisfy the requirements for the intended service conditions.

There are a number of standards throughout the world that relate to weld symbols, but in the United States, AWS A2.4, Standard Symbols for Welding, Brazing, and Non-destructive Examination, is the standard a

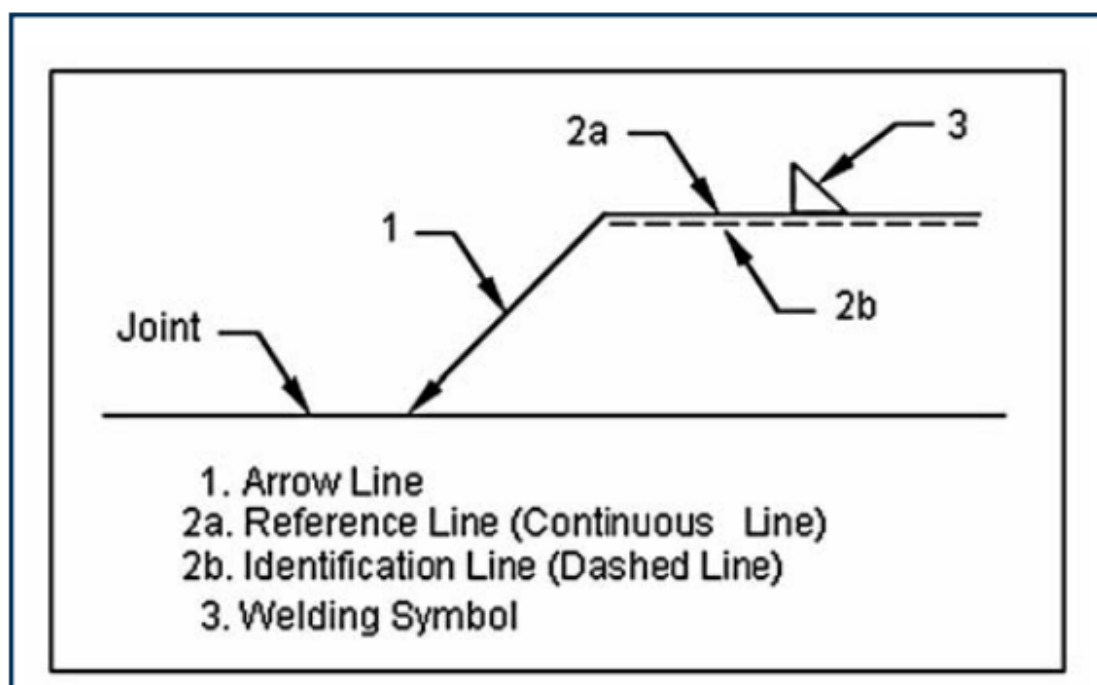
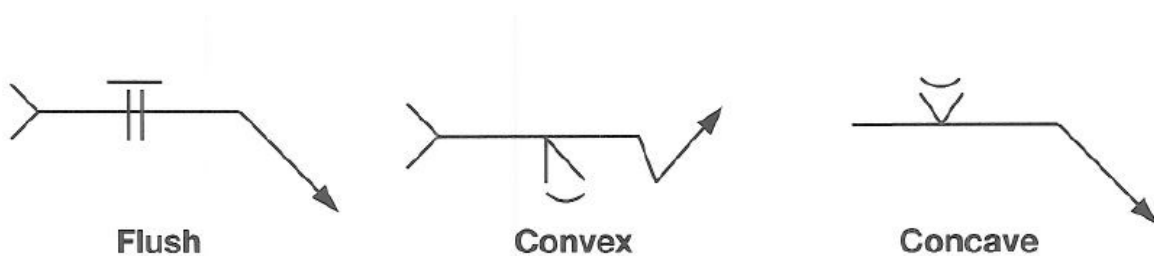
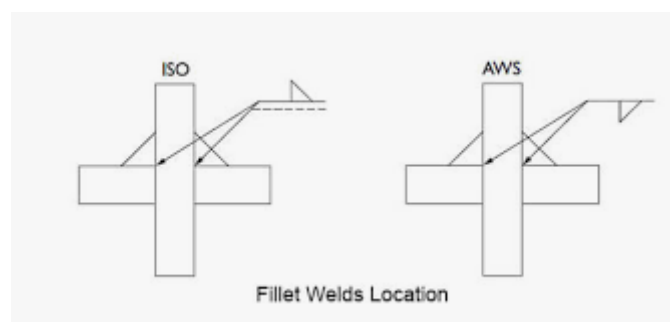


Fig. 19 — ISO welding symbol system.

Desired Weld	ISO 2553	AWS A2.4

Fig. 20 — Comparison of ISO and AWS fillet welding symbols.



There are three types of contours for welds

8. Determine the finishing method.

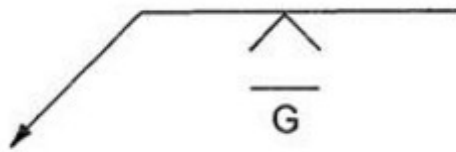
The welding symbol can also tell you how the contour is to be achieved. For example, the requirement may be a flat contour by grinding. **AWS A2.4:2007 – Standard Symbols for Welding, Brazing, and Non-destructive Examination**

C – Chipping
G – Grinding
H- Hammering
M – Machining

P – Planishing
R – Rolling
U – Unspecified

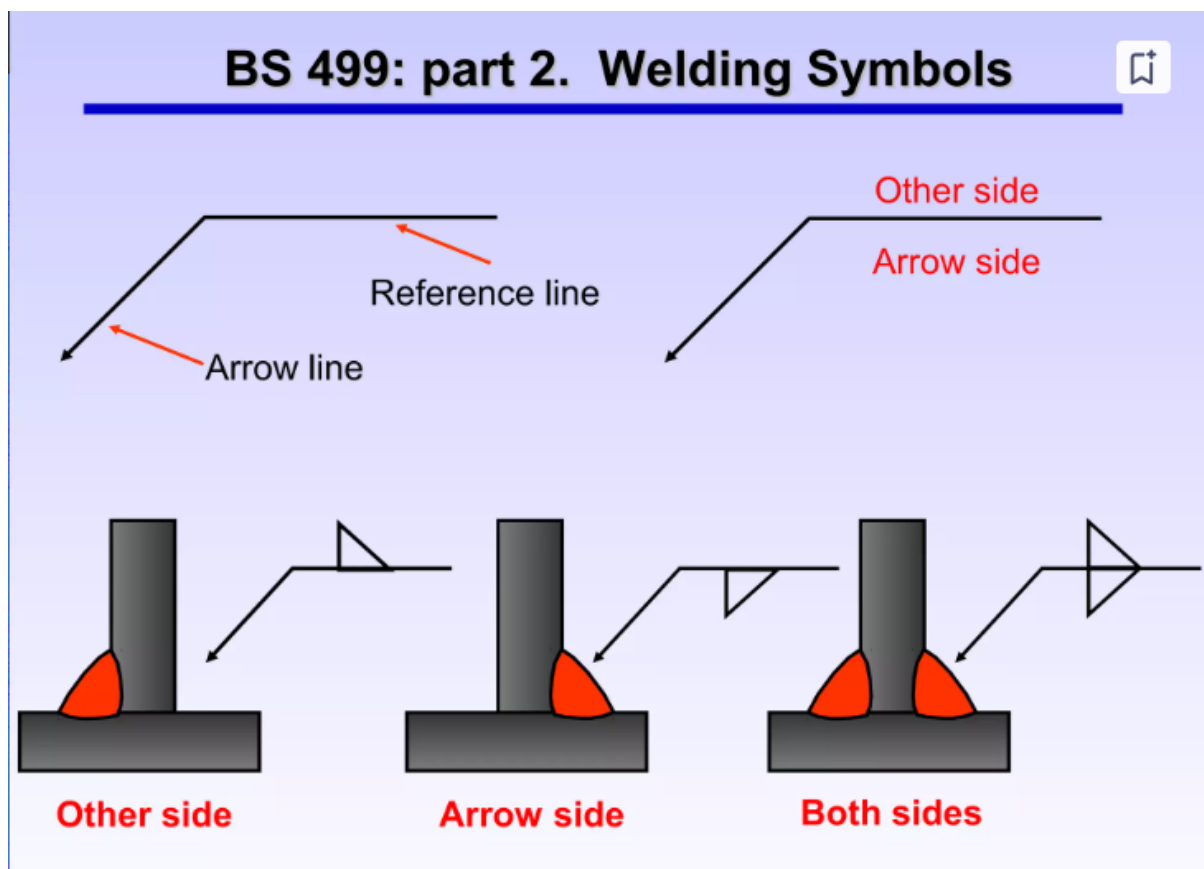
specifies finishing methods with a 7th being “unspecified method:”

These notations will appear above (or below) the contour symbol as seen below where a flat contour is achieved by grinding.

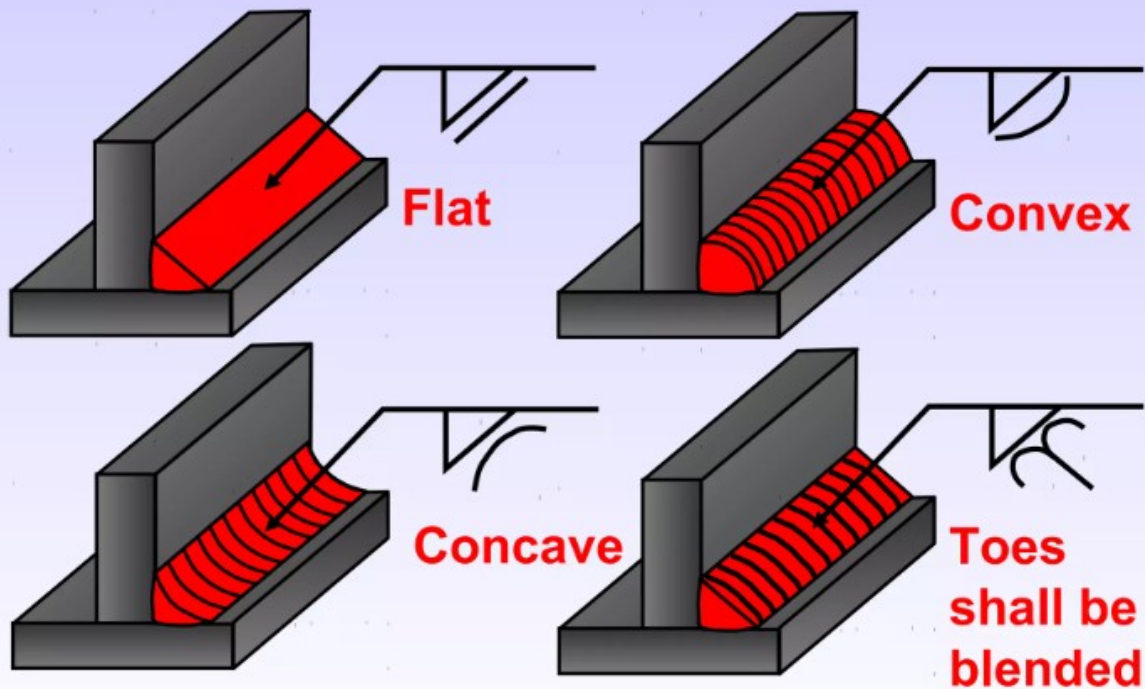
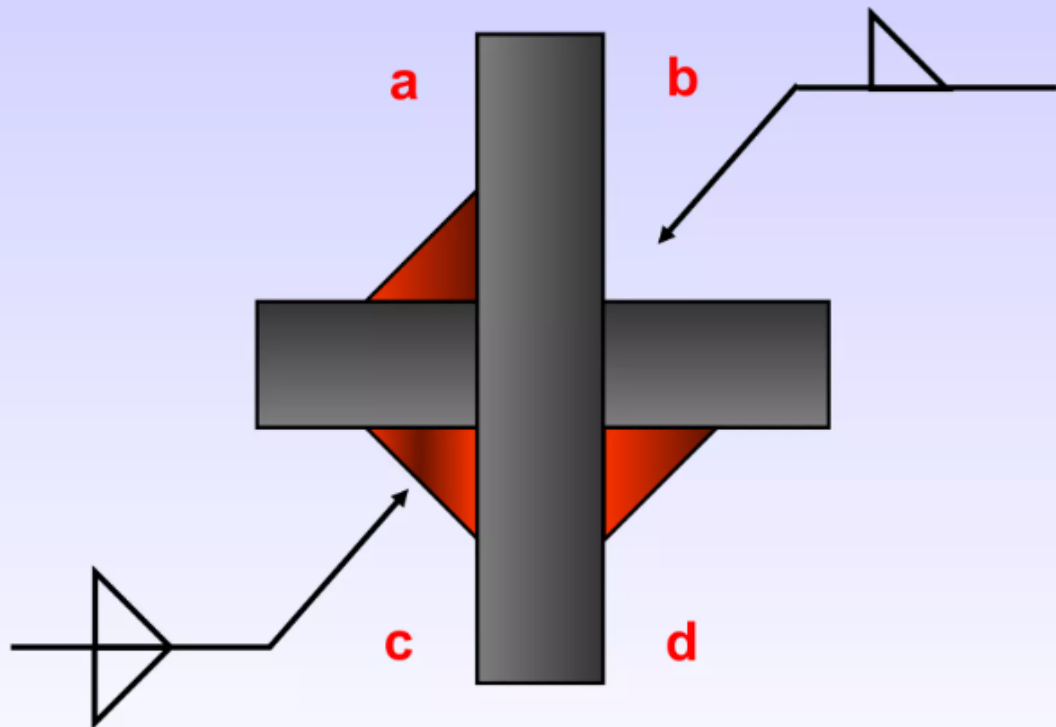


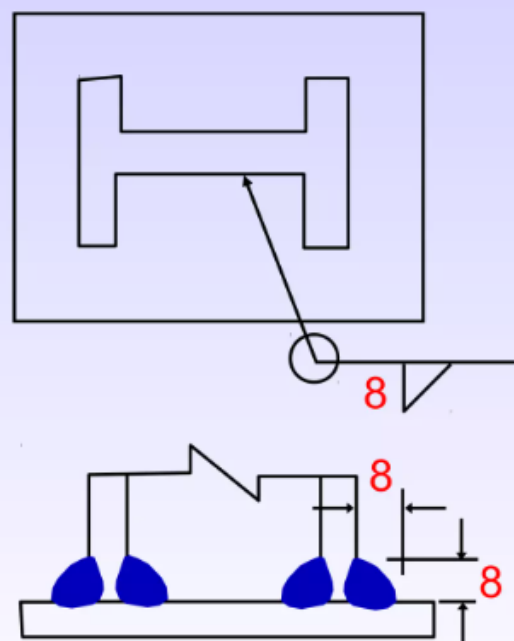
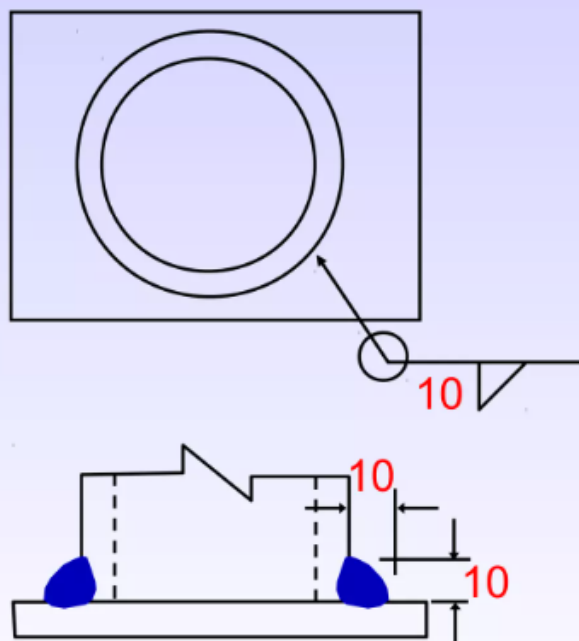
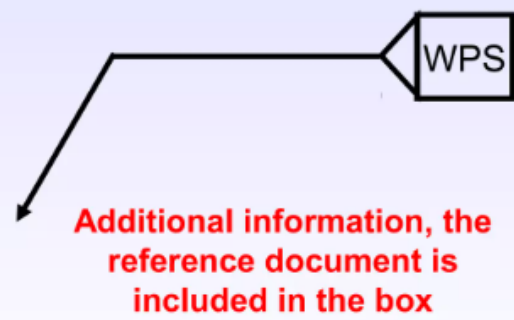
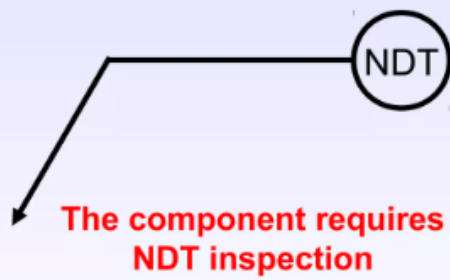
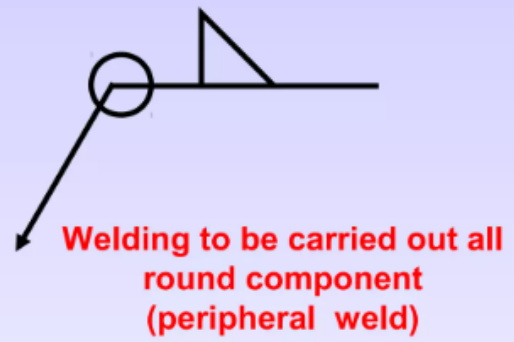
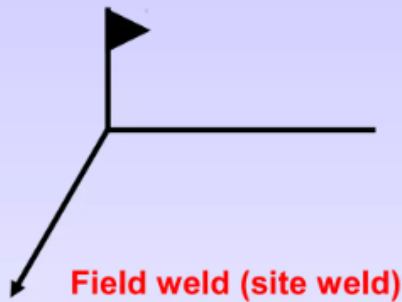
The symbol calls for a groove weld with a flat finish achieved by grinding.

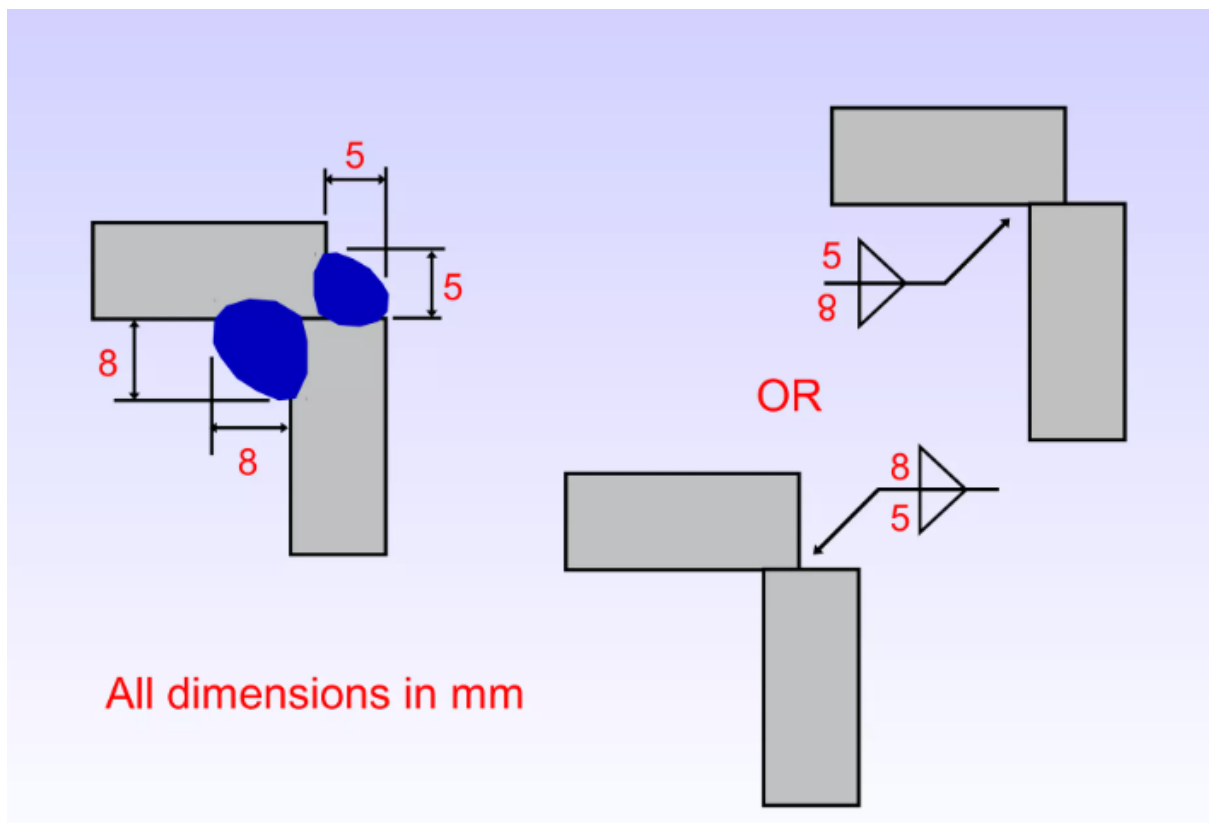
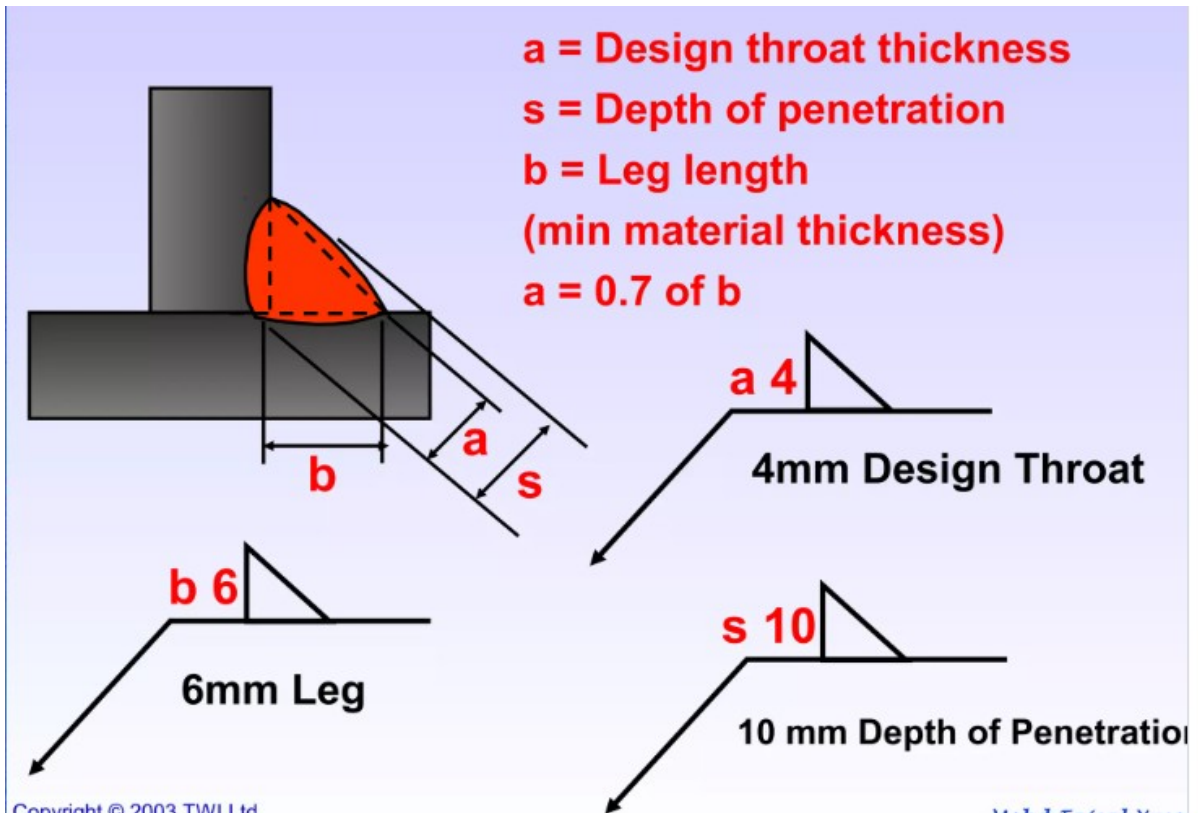
<https://weldinganswers.com/9-basic-steps-to-reading-welding-symbols-for-groove-welds/>

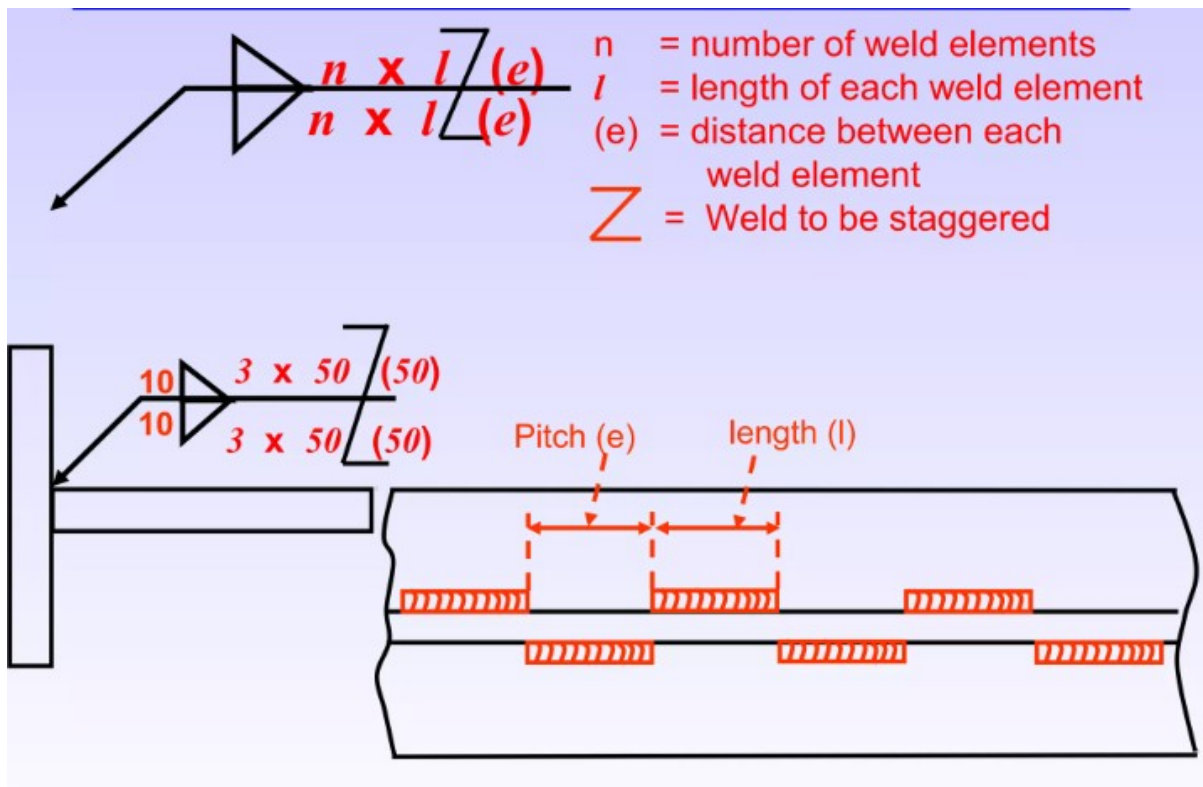
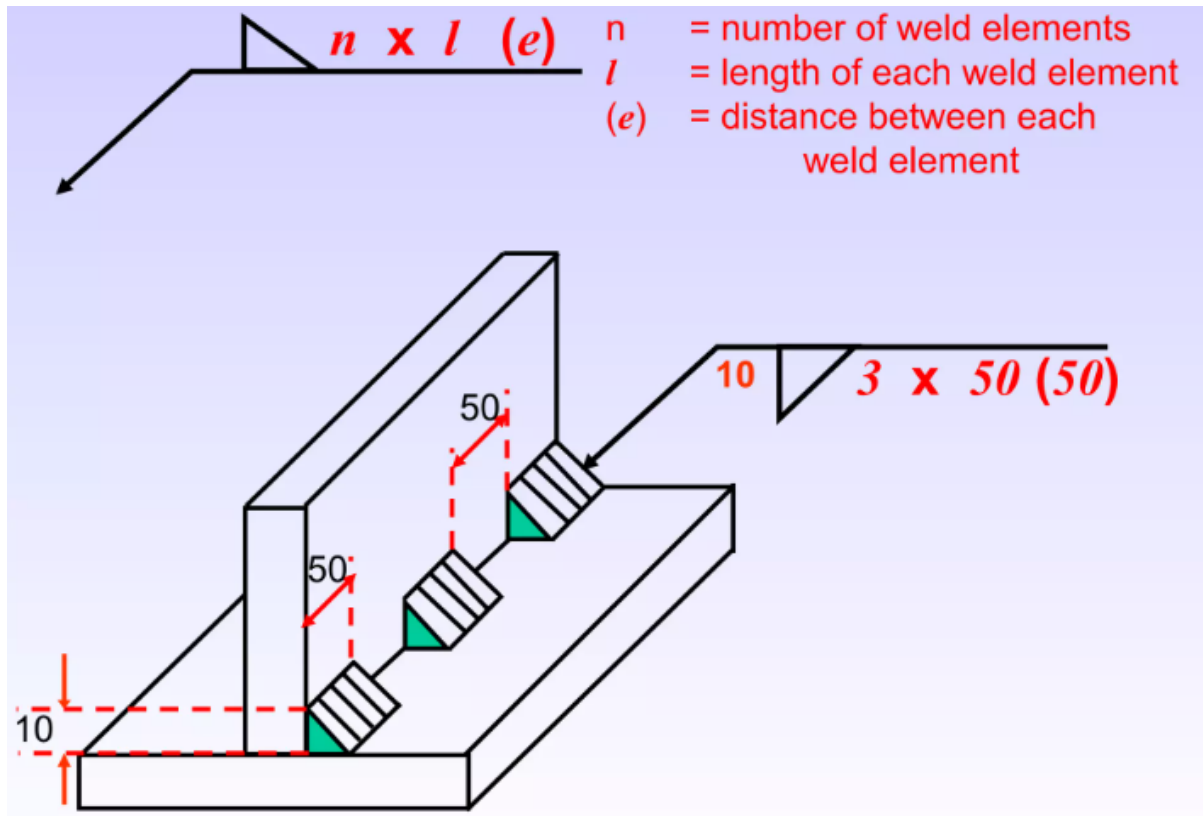


BS 499: part 2. Welding Symbols

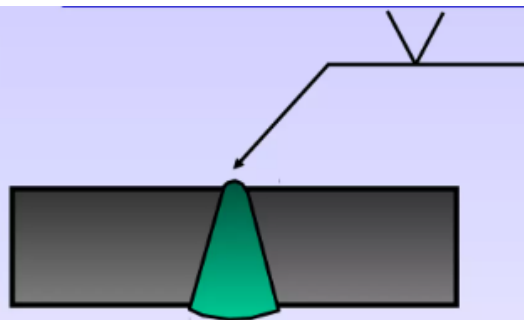
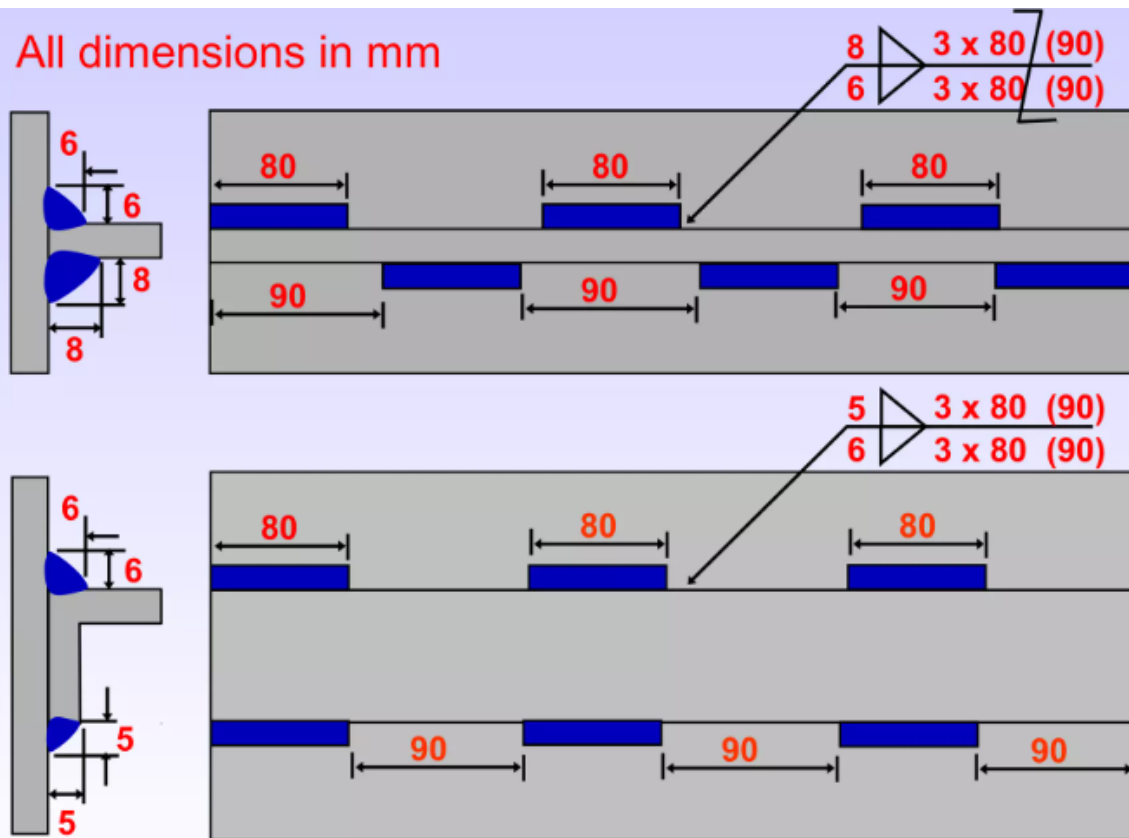




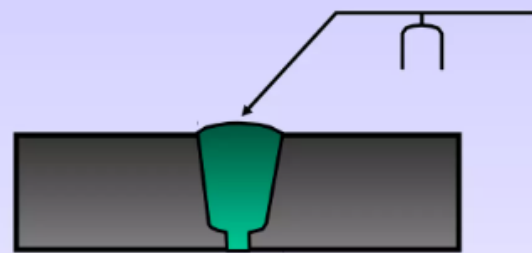




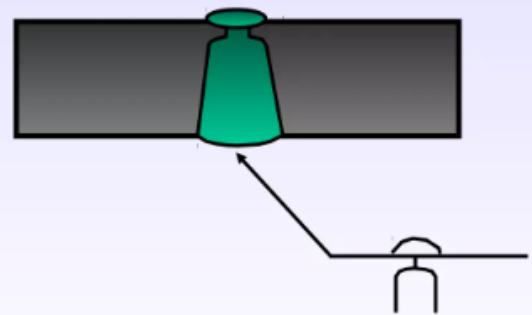
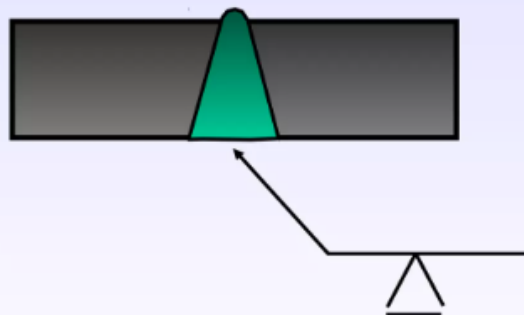
All dimensions in mm

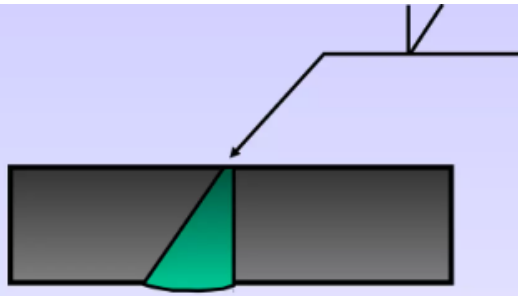


Single-V Butt

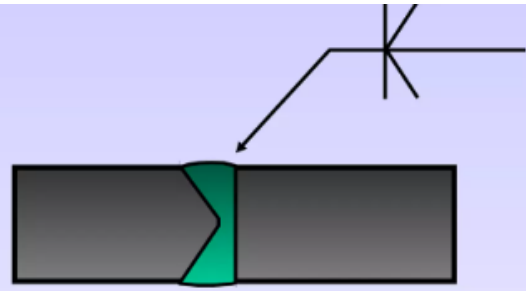


Single-U Butt

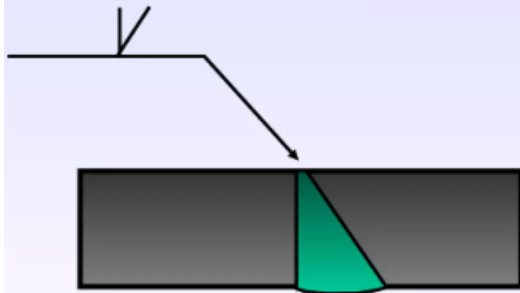




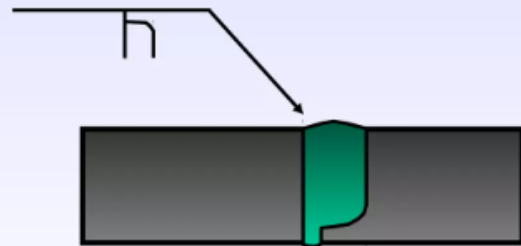
Single-bevel butt



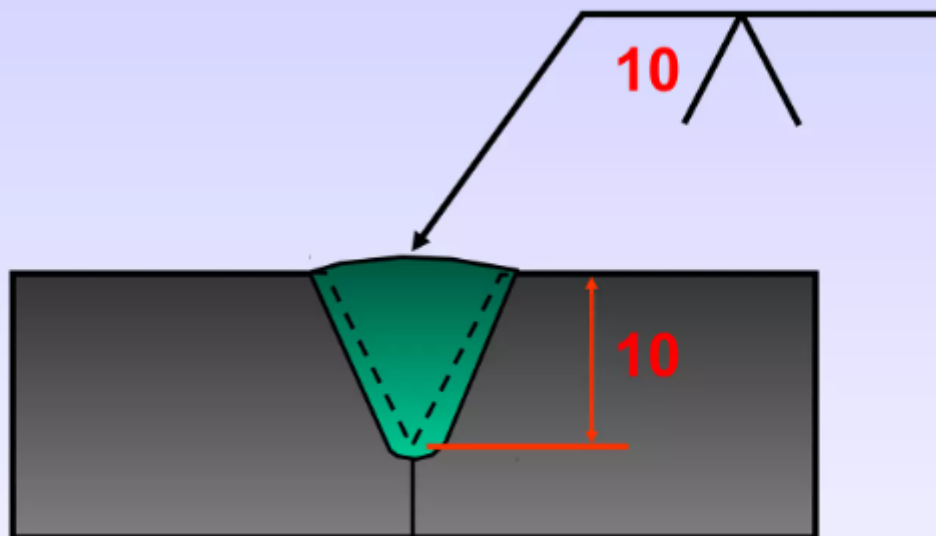
Double-bevel butt



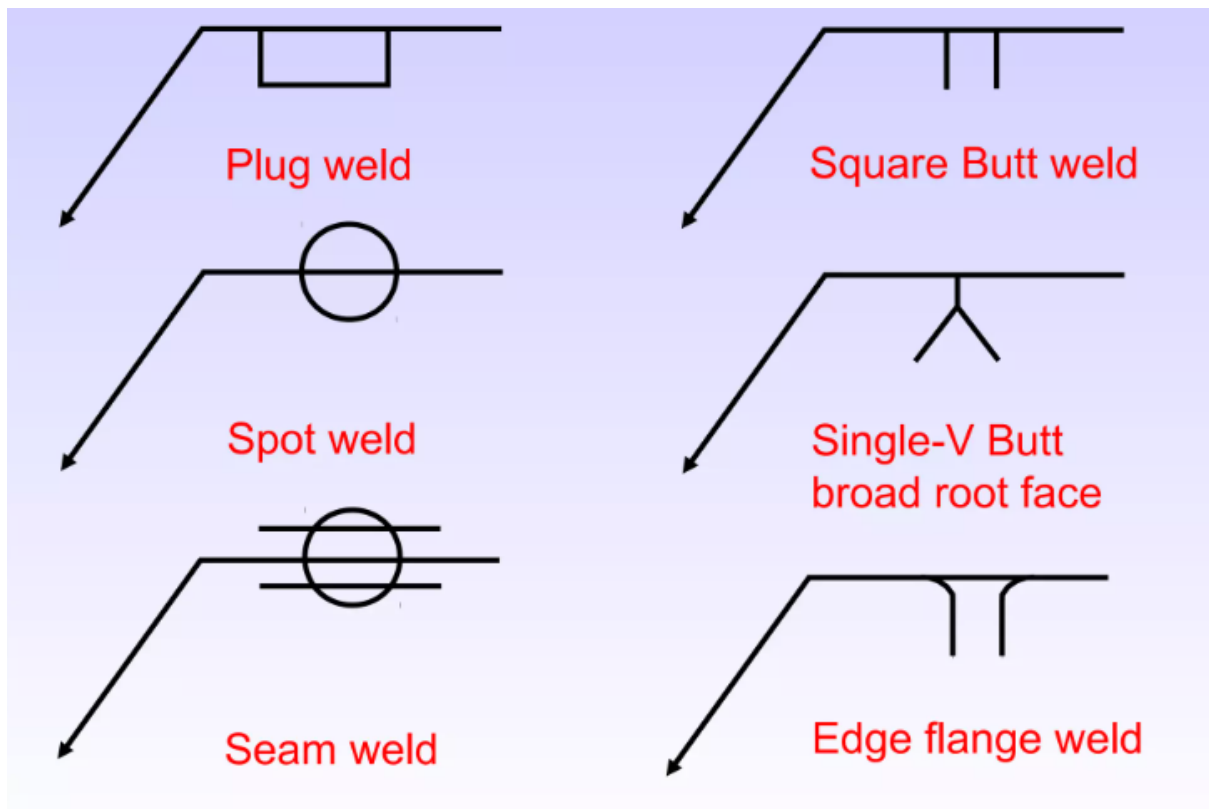
Single-bevel butt



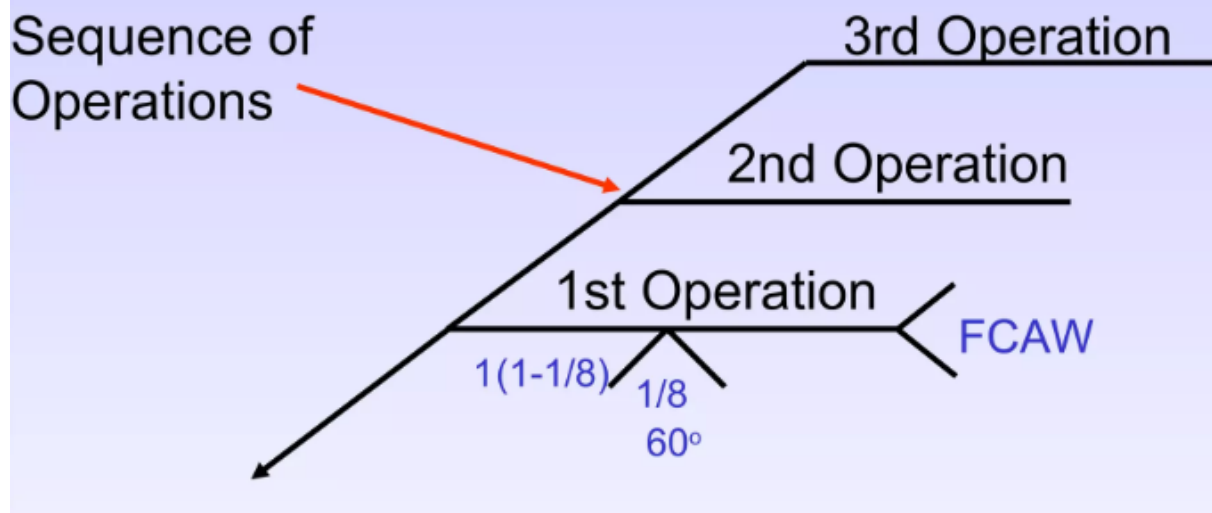
Single-J butt



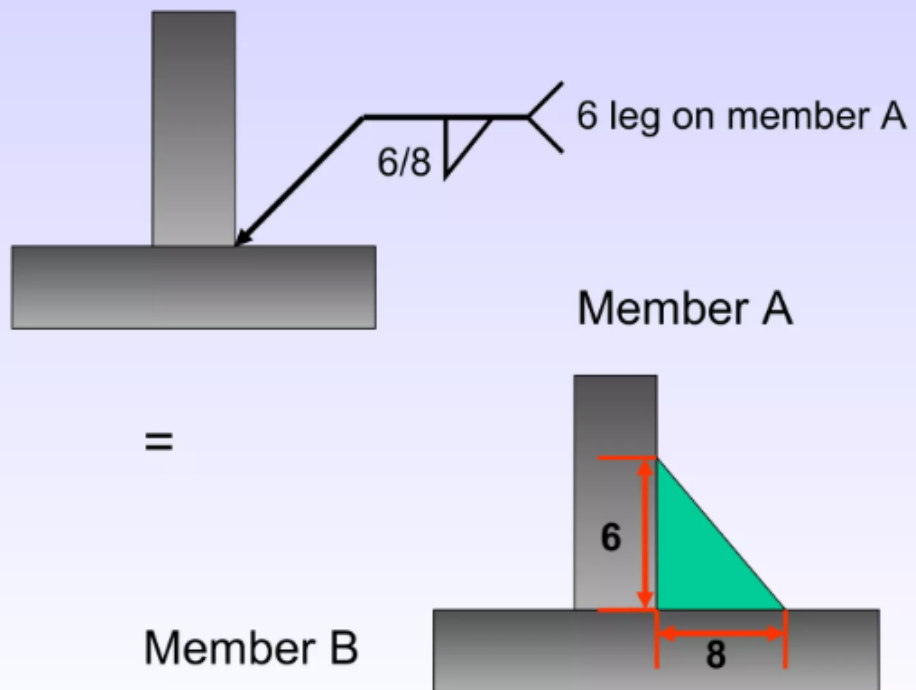
Partial penetration single-V butt



AWS. Welding Symbols



Dimensions- Leg Length

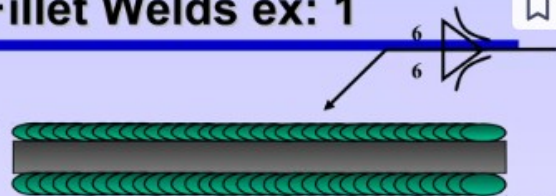


Any Questions?

Rectangular Snip

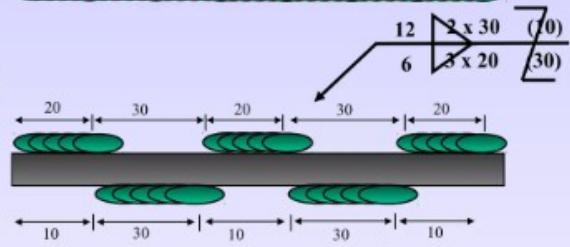
Ans BS 499: prt. 2 Fillet Welds ex: 1

Welded both sides: A continuous concave fillet weld 6mm leg lengths.

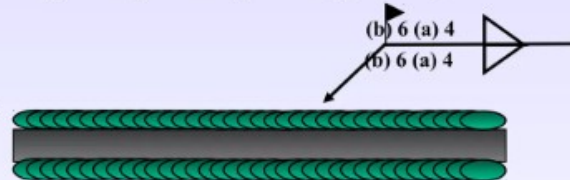


Welded arrow side: Three intermittent fillet welds, 6 mm leg lengths, the length of each weld 20 mm, the distance between each weld 30 mm.

Welded other side: Two intermittent fillet welds 12 mm leg lengths, the length of each weld 30 mm, the distance between each weld 10 mm. Welds to be staggered.

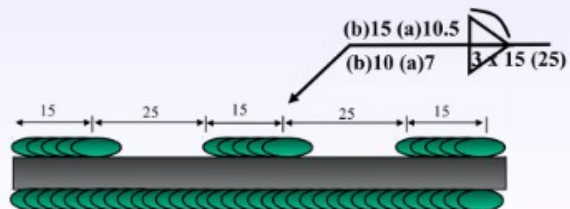


Welded both sides: A continuous fillet weld, 6 mm leg lengths, 4 mm throat thickness, welds to be carried out on site.



Welded arrow side: Three intermittent fillet welds 10 mm leg lengths, 7mm throat thickness, length of each weld 15 mm, the distance between each weld 25 mm.

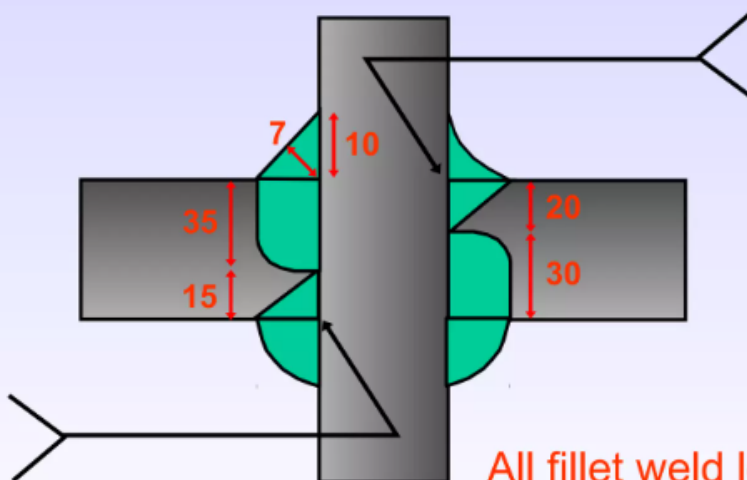
Welded other side: A continuous convex fillet weld, 15 mm leg length, 10.5 mm throat thickness.



BS 499: prt. 2 Butt Weld ex: 1

Complete the symbol drawing for the welded cruciform joint provided below

All welds are welded with the MIG process and fillet welds with the MMA process



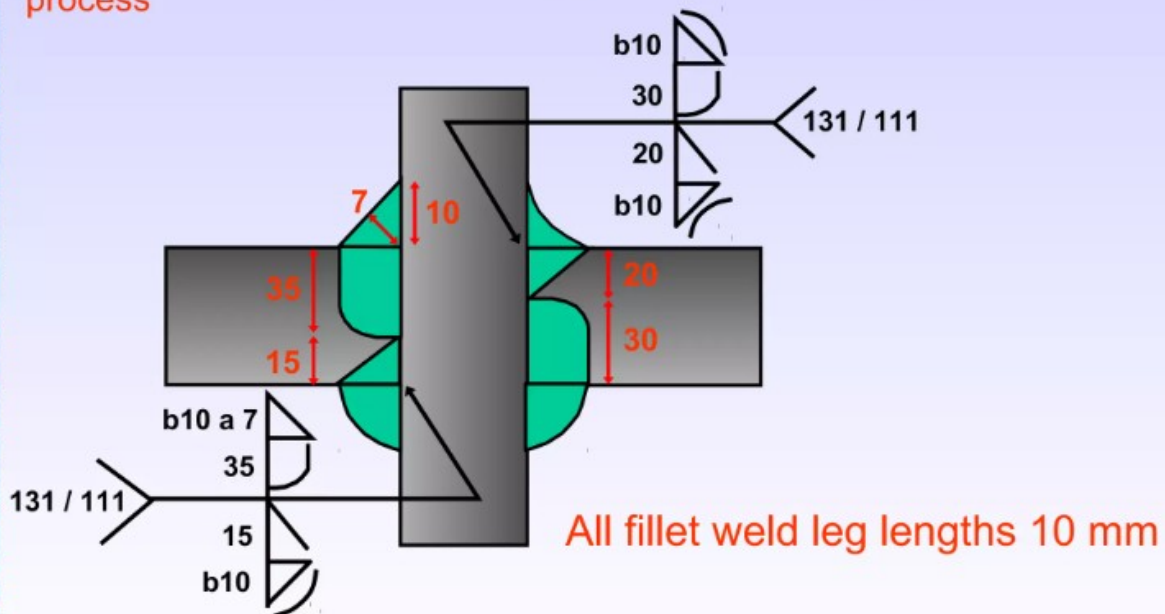
All fillet weld leg lengths 10 mm

Ans BS 499: prt. 2 Butt Weld ex: 1



Complete the symbol drawing for the welded cruciform joint provided below

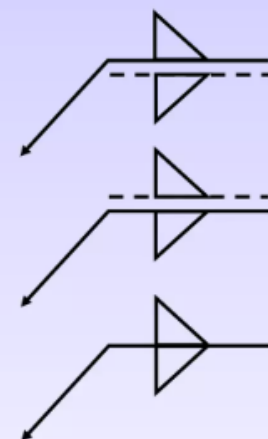
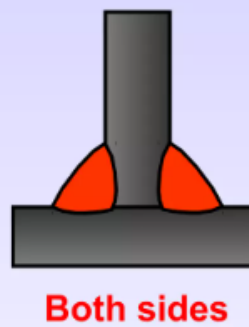
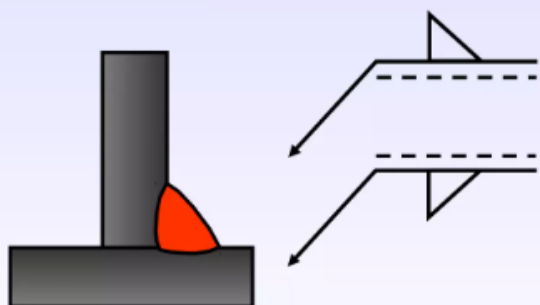
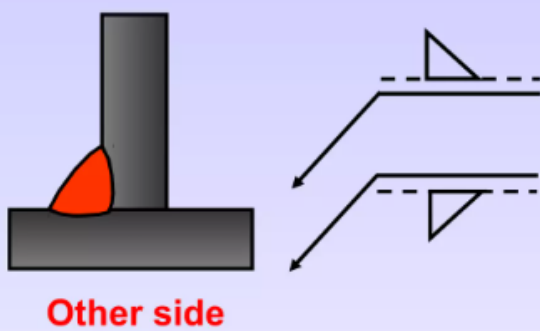
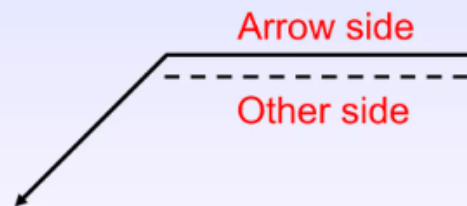
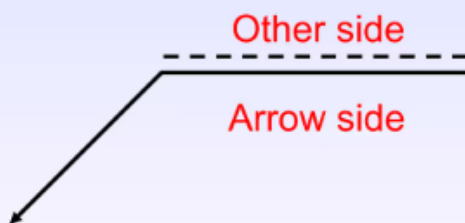
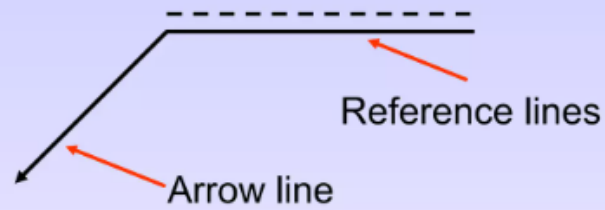
All welds are welded with the MIG process and fillet welds with the MMA process

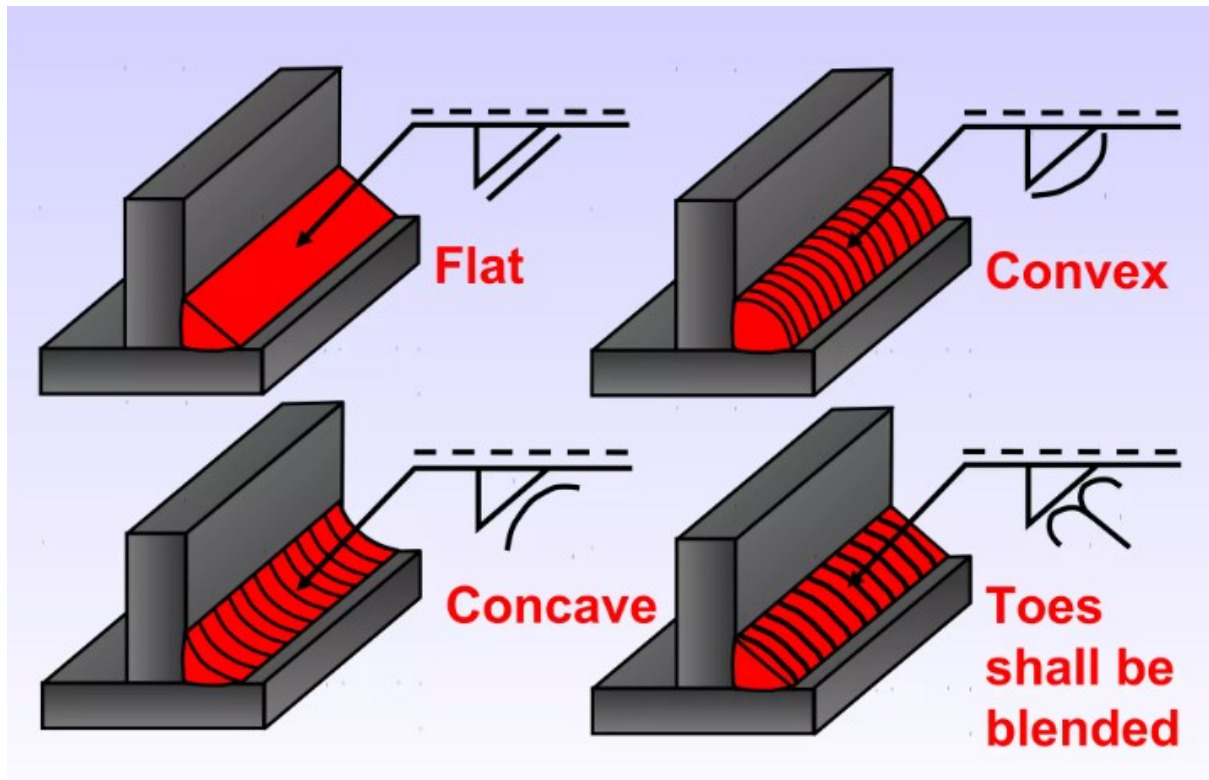


BS EN 22553 (ISO 2553) Welding Symbols

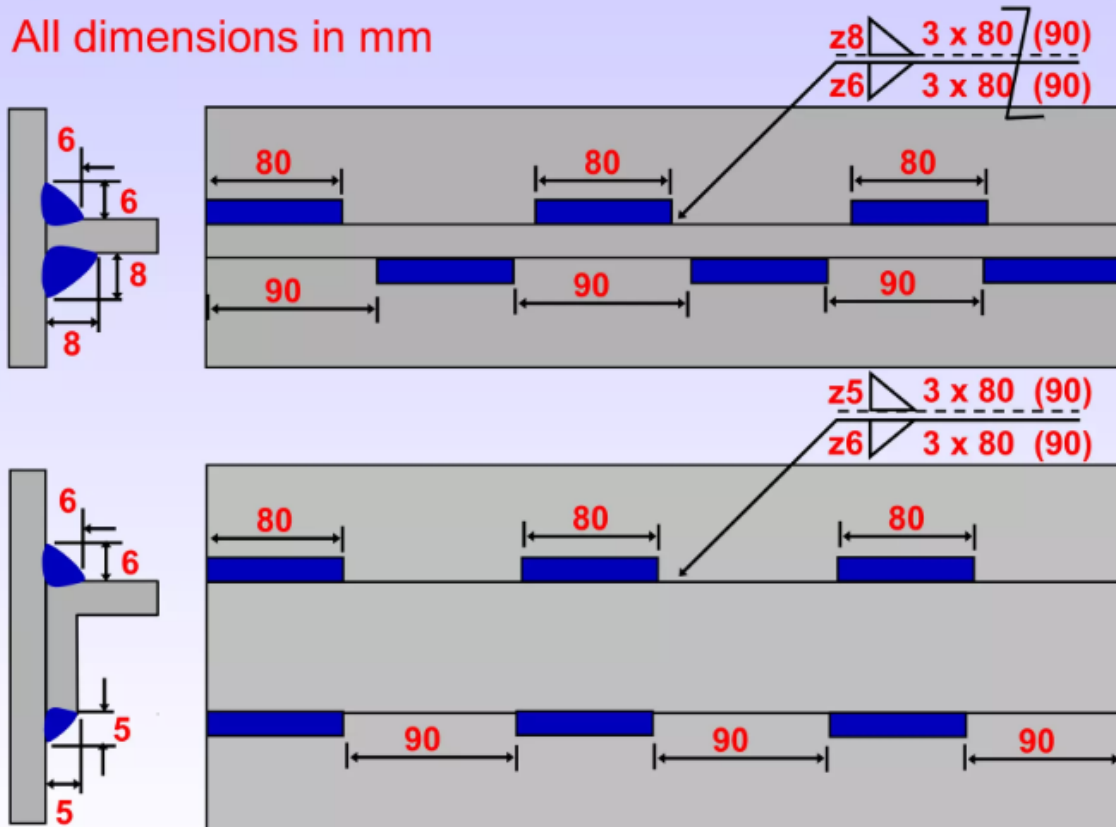
NB: WE NOT USING ISO 2553 WELDING SYMBOLS HERE WE DOING THEM JUST FOR YOU TO UNDERSTAND THE DIFFERENCE BETWEEN ISO AND AWS

BS EN 22553 (ISO 2553) Welding Symbols





All dimensions in mm



Any Questions?

The primary element of any welding symbol is referred to as the

Reference line

The term pitch refers to the spacing of intermittent fillet weld

Centre to center

The weld joint formed when two parallel members overlap each other is called

Lap joint